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ON ENDOGENOUS COMPETITIVE BUSINESS CYCLES

BY JEAN-MICHEL GRANDMONT¹

This paper develops an example in which persistent deterministic business cycles appear in a purely endogenous fashion under *laissez-faire*. These cycles are not attributable to exogenous “shocks” nor to any variation of policy since there are none in the model. Markets clear in the Walrasian sense at every date, and traders have perfect foresight along the cycles.

The origin of these cycles is the potential conflict between the wealth effect and the intertemporal substitution effect that are associated with real interest rate movements. Business cycles appear in particular when the degree of concavity of a trader’s utility function is sufficiently higher for old agents than for younger ones. The techniques employed to study the occurrence and the stability of such business cycles are borrowed partly from recent mathematical theories that have been constructed by using the notion of the “bifurcation” of a dynamical system in order to explain the emergence of cycles and the transition to turbulent (“chaotic”) behavior in physical, biological, or ecological systems.

The equilibrium level of output is shown to be negatively related to the equilibrium level of the real interest rate. A similar relation exists (but in the opposite direction) between equilibrium real money balances and real interest rates. These relations hold both in the long run, i.e. along business cycles, and in the short run, i.e. on the transition path, and whether movements of the real interest rate are anticipated or not. The basic ingredient there is the condition that older agents have a higher marginal propensity to consume leisure.

Monetary policy by means of nominal interest payments is shown to be extremely effective. A permanent change of the rate of growth of the money supply by these means is superneutral. Yet, there exists a very simple deterministic countercyclical policy that enables monetary authorities to stabilize completely business cycles and to force the economy back to the unique (Golden rule) stationary state. Due to the nonlinearity of the model such a policy affects not only the variances of real equilibrium magnitudes but also their means.

INTRODUCTION

THE BELIEF THAT THE LONG RUN EQUILIBRIUM of a competitive monetary economy that does not experience any exogenous shocks—whether originating from the external environment or from policy—should be modeled as a state that

¹ The material of this paper has been presented as part of the Walras–Bowley lecture at the North American Summer Meetings of the Econometric Society, Stanford, June, 1984. I wish to thank very much the members of the ensuing discussion panel, K. Arrow, R. Hall, R. Lucas, C. Sims, and J. Stiglitz for their perceptive comments. An earlier version was presented at the Workshop on Price Adjustment, Quantity Adjustment, and Business Cycles, in October, 1983, at the Institute for Mathematics and Its Applications of the University of Minnesota, organized by Hugo Sonnenschein. Special thanks are due to the discussants of the paper at the Workshop, Daniel Goroff, José Scheinkman, Christopher Sims, Neil Wallace, and Mike Woodford, for their valuable comments and constructive criticisms.

I wish to thank very much Rose Anne Dana and Pierre Malgrange who introduced me to the mathematics of the subject: without them, this research could not have been completed. Christian Gouriéroux deserves also a special mention for his decisive advice in connection with the stability issue of Section 3. Rose Anne Dana and Dominique Lévy ran the computer experiment reported in Section 4. Their help is gratefully acknowledged. Earlier simulations by Alain Morineau were also very valuable.

I had stimulating conversations with and received useful comments from many colleagues at various stages of this work. I wish to thank them all, and especially Philippe Aghion, Albert Ando,

is stationary or perhaps growing at a constant rate seems to be deeply rooted in the mind of some economists.

The most outspoken believers in the market's invisible hand go indeed as far as claiming that any departure from a long run Walrasian equilibrium should be regarded as purely transitory and that accordingly the basic tendencies of a competitive economy may be represented adequately by such a "Classical" stationary equilibrium. The most recent reformulation of the Classical approach has been to model economic fluctuations by adding random shocks to the deterministic stationary state and to underscore the role of incomplete (and asymmetric) information in the influence of economic policy on real equilibrium variables. The outcome of this reformulation is a model that preserves very cleverly stationarity while incorporating in the analysis something that looks like business cycles (Barro [5], Kydland and Prescott [40], Lucas [42, 43, 44, 45, 46], McCallum [47], Sargent and Wallace [50]).² An important implication of many, but not all, of these models is that the systematic (deterministic) component of economic policy can have no real effect whenever it is anticipated by the private sector.

The arguments put forward by the opposing (Keynesian) school appear often, by contrast, almost exclusively defensive. Proponents of this school seem to accept in effect the *theoretical* validity of the claim according to which the long run equilibrium positions of a competitive economy may be described by deterministic stationary states. They tend to question primarily the *practical* relevance, for the description of short run and medium run phenomena, of the mere notion of a long run stationary equilibrium and of its underlying assumptions. The list is long: prices cannot move fast enough to clear markets, anticipations adjust only slowly, New Classical macroeconomic models rely upon extremely specific assumptions concerning the distribution of information, the Classical stationary state may be unstable or convergence to it may be so slow that it becomes practically irrelevant in calendar time, and so on.³

The purpose of this work is to demonstrate that, by contrast to currently accepted views, a competitive monetary economy of which the environment is stationary may undergo persistent and large deterministic fluctuations under *laissez faire*; that these cyclical fluctuations may display moreover the sort of correlations that recent Classical macroeconomic models have sought to incorporate, even under complete information, without having to make the *ad hoc* assumption that cycles are due to exogenous shocks; and finally, that the Government,

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Last, but not least, I wish to express my gratitude to the four referees, in particular to Margaret Bray, for their very professional reports and their suggestions, which led to significant improvements of the paper. Financial support from the French Commissariat General du Plan, the University of Lausanne, and the U.S. Office of Naval Research under Contract ONR-R00014-79-C-0685 at the Institute for Mathematical Studies in the Social Sciences at Stanford University is also gratefully acknowledged.

² See also Grandmont and Hildenbrand [35].

³ For an excellent account of the long feud between Keynesian and (Old, Neo- or New) Classical economists, see e.g. Tobin [58].

in the face of such autonomous deterministic fluctuations, has indeed in principle the power to stabilize the economy by implementing simple deterministic—and publicly known—countercyclical policies.

Although one of the goals of the present work is to develop concepts and methods that can be applied, it is hoped, to a larger class of situations, the analysis will proceed by studying a particular example, i.e. an overlapping generations model very much like the model developed by R. J. Lucas in his seminal paper [42], with the noticeable difference that we shall assume that the economy is *not* subjected to any shock of any sort. Business deterministic cycles will be shown to appear in a purely endogenous fashion under *laissez faire*. Markets will be assumed to clear in the Walrasian sense at every date, and traders will have perfect foresight along the cycles.

The origin of these endogenous deterministic cycles will be seen to be the potential conflict between the wealth effect and the intertemporal substitution effect that are associated with real interest rate movements. Business cycles will emerge in particular when the degree of concavity of a trader's utility function—which we shall measure, although there is no uncertainty in the model, by the so-called Arrow–Pratt “relative degree of risk aversion”—is sufficiently higher for old agents than for younger ones.⁴ An important outcome of the analysis will be that cycles of different periods will typically coexist—in some cases, there may be an infinity of these. The techniques employed to study the occurrence and the stability of such business cycles will be borrowed partly from recent mathematical theories that have been constructed by using the notion of the “bifurcation” of a dynamical system in order to explain the emergence of cycles and the transition to turbulent (“chaotic”) behavior in physical, biological, or ecological systems.⁵

The equilibrium level output will be shown to be negatively related to the equilibrium level of the real interest rate. A similar relation exists (but in the opposite direction) between equilibrium real money balances and real interest rates. These relations hold both in the long run, i.e. along business cycles, and in the short run, i.e. on the transition path, and whether movements of the real

⁴ The idea that endogenous deterministic cycles may emerge in an overlapping generations model is already present in the literature. For instance David Gale [30] presents a numerical example of a cycle of period 2, while David Cass [12], Brock and Scheinkman [11] discuss the possibility of their occurrence. Moreover, the deterministic business cycles that are the subject of this paper may be assimilated to what has been called recently “sunspots” equilibria. The analysis of sunspot equilibria has been started by Karl Shell [51] and later developed by C. Azariadis [2], D. Cass and K. Shell [13, 14], and Spear [53]. Independently of the present work, C. Azariadis and R. Guesnerie [4] establish in a model similar to ours that there are sunspots equilibria if and only if deterministic cycles exist.

A recent paper by P. Diamond and D. Fudenberg [20] provides an example of an endogenous rational expectations business cycle in a search equilibrium model, in an otherwise stationary environment. Finally, Benhabib and Nishimura [8, 9] investigate the occurrence of endogenous deterministic cycles in a neoclassical growth model.

⁵ One important mathematical reference in this field is Collet and Eckmann [16]. For a stimulating review of various applications of the theory, see May [48]. Part of this theory has been already applied in economics or game theory, in particular by Benhabib and Day [6, 7], Dana and Malgrange [17], Day [18, 19], Jensen and Urban [39], Rand [49]. Some results of this theory that seemed (to me) relevant and useable by economic theorists are reviewed in Grandmont [34].

interest rate are anticipated or not. The basic ingredient there will be the condition that older agents have a higher marginal propensity to consume leisure.

Finally, monetary policy by means of nominal interest payments will be shown to be extremely effective. A permanent change of the rate of growth of the money supply by these means will be seen to be superneutral. Yet, it will be shown that there exists a very simple deterministic counter-cyclical policy that enables monetary authorities to stabilize completely business cycles and to force the economy back to the unique (Golden rule) stationary state. Due to the nonlinearity of the model, such a policy affects not only the variances of real equilibrium magnitudes but also their means. The central point here is that there are typically many long run periodic equilibria that coexist under *laissez faire*, and that policies may be designed which force the economy to settle at only one of these—here the stationary state.⁶

The paper is organized as follows. We specify in Section 1 the structure of the model and study there the traders' microeconomic behavior. Long run periodic equilibria with perfect foresight are defined in Section 2. The issue of the stability of these periodic equilibria is partly analyzed in Section 3. The existence, the multiplicity, and the bifurcations of periodic competitive equilibria are investigated systematically in Section 4. The long run and short run relationships between equilibrium output or real balances, and anticipated or unanticipated real interest rates are established in Section 5. Finally, the impact of monetary policy through deterministic proportional money transfers is dealt with in Section 6. A few concluding remarks are given in Section 7, while some proofs are gathered in a separate Appendix.

Section 4 is the most technical, and although it is in some respects the most interesting one, the nonmathematically oriented reader may skip it on a first reading. Section 5 and Section 6 can be read right after the first two sections of the paper.

1. BEHAVIORAL ASSUMPTIONS

We shall use the simple structure of an overlapping generation model, with a constant population and without bequests, in which agents live two periods only. For simplicity we shall assume that all agents are identical, or equivalently that there is a single member in each generation. There will be accordingly two agents in every period, one “young” and one “old.” The model involves one perishable consumption good, which is produced from the labor that is supplied by consumers. There is no production lag, and producing one unit of output requires one unit of labor. Young consumers have the opportunity to save part of their income in each period by holding a nonnegative money balance. For the most

⁶ The idea that there may be a large number of “rational expectations” or perfect foresight equilibria in a monetary economy and that accordingly, one possible role of policy is to select one of these, is also already present in the literature. In particular, this fact has been well known by theorists who worked with the overlapping generations model; see e.g. Geanakoplos and Polemarchakis [31]. The point has been most forcefully reiterated recently by F. Hahn [38].

part of the paper, the money stock will be assumed to be constant over time. It will be denoted $M > 0$.

At each date t , there are competitive spot markets for the consumption good, for labor, and for money. The money price of the good and the money wage rate need not be distinguished, however, since profit maximization requires that the equilibrium real wage be equal to unity.

An agent's intertemporal characteristics may then be described as follows. Consumption c_τ in each period τ of his life ($\tau = 1, 2$) must be nonnegative. On the other hand, it is assumed that the agent has a labor endowment in each period of his life, l_τ^* , and that his labor supply l_τ or equivalently his consumption of leisure $l_\tau^* - l_\tau$ must satisfy $0 \leq l_\tau^* - l_\tau \leq l_\tau^*$ for $\tau = 1, 2$. The agent's intertemporal tastes are represented by the separable utility function $U_1(c_1, l_1^* - l_1) + U_2(c_2, l_2^* - l_2)$ which is defined on the set of c_τ and $l_\tau^* - l_\tau$ that satisfy the foregoing feasibility constraints. We shall assume:

ASSUMPTION (1.a): $l_1^* > 0$ and $l_2^* > 0$.

ASSUMPTION (1.b): $U_\tau(c_\tau, l_\tau^* - l_\tau)$ is continuous, increasing in each argument, and strictly concave for $\tau = 1, 2$.

We consider now the decision problem that a young agent has to solve at an arbitrary date. Let $p > 0$ be the money price of the good that he observes in the current period and let $p^e > 0$ be the price that he expects for the next date (current and expected money wages are taken to be equal to p and p^e respectively). The agent's problem is then to choose his current consumption $c_1 \geq 0$, his current labor supply $0 \leq l_1 \leq l_1^*$, his demand for nominal money $m \geq 0$, and to plan for the next date his future consumption $c_2 \geq 0$ and labor supply $0 \leq l_2 \leq l_2^*$, so as to maximize his intertemporal utility function subject to the current and expected budget constraints

$$(1.1) \quad p(c_1 - l_1) + m = 0 \quad \text{and} \quad p^e(c_2 - l_2) = m.$$

Under Assumptions (1.a), (1.b), the problem has a unique solution. The optimum value of $(c_\tau - l_\tau)$ depends only on $\theta = p/p^e$, or equivalently on the consumer's expected real interest rate $\theta - 1$, and is denoted $z_\tau(\theta)$, for $\tau = 1, 2$ (absence of money illusion). Since one unit of labor yields one unit of good, these values can be interpreted as the trader's current and expected *excess demands* for the good. On the other hand, the agent's *demand for money* $m^d(p, p^e)$ is given by

$$(1.2) \quad m^d(p, p^e) \equiv -pz_1(\theta) \equiv p^ez_2(\theta)$$

which implies of course

$$(1.3) \quad \theta z_1(\theta) + z_2(\theta) \equiv 0 \quad \text{for every } \theta > 0.$$

It is convenient to decompose the trader's decision problem into two subproblems as follows. First, given $a_\tau \geq 0$, the agent maximizes $U_\tau(c_\tau, l_\tau^* - l_\tau)$ subject to

$$(1.4) \quad c_\tau + (l_\tau^* - l_\tau) = a_\tau$$

Under Assumptions (1.a), (1.b), this problem has a unique solution which determines the agent's consumption $c_\tau(a_\tau)$ and his labor supply $l_\tau(a_\tau)$ in each period of his life as a function of his "real wealth" a_τ for $\tau = 1, 2$.

Let $V_\tau(a_\tau)$ be the maximum of U_τ subject to (1.4) and consider the problem of maximizing $V_1(a_1) + V_2(a_2)$ with respect to $a_1 \geq 0$, $a_2 \geq 0$, $m \geq 0$ subject to

$$(1.5) \quad pa_1 + m = pl_1^* \quad \text{and} \quad p^e a_1 = p^e l_2^* + m.$$

By comparing (1.4) and (1.5) with (1.1), it is clear that the optimum values of $a_\tau - l_\tau^*$ and of m that result from this last problem coincide indeed with $z_\tau(\theta)$ and $m^d(p, p^e)$.

The argument shows that considering a variable production model like this one is formally equivalent to looking at an exchange economy (i.e. without production) in which every agent is endowed with the quantity l_τ^* of the good in each period of his life. The procedure will also allow us to state our assumptions more compactly on the indirect utility functions V_τ instead of deriving them for the original function U_τ .

Under Assumptions (1.a), (1.b), each V_τ is continuous, strictly concave and increasing. We shall make in fact the stronger assumption:

ASSUMPTION (1.c): *For each $\tau = 1, 2$, the indirect utility function V_τ is continuous on $[0, +\infty)$ and twice continuously differentiable on $(0, +\infty)$, with $V'_\tau(a_\tau) > 0$, $\lim_{a_\tau \rightarrow 0} V'_\tau(a_\tau) = +\infty$, $V''_\tau(a_\tau) < 0$.*

We end up this section with a brief analysis of a few elementary facts about the excess demand functions z_1 and z_2 that will be used repeatedly in the sequel.

LEMMA 1.1: *Assume (1.a) and (1.c) and let $\bar{\theta} = V'_1(l_1^*)/V'_2(l_2^*)$. Then $z_1(\theta)$ and $z_2(\theta)$ are continuous on the open interval $(0, +\infty)$. Moreover, (i) $z_1(\theta) = z_2(\theta) = 0$ whenever $\theta \leq \bar{\theta}$, and $-l_1^* < z_1(\theta) < 0$, $z_2(\theta) > 0$ whenever $\theta > \bar{\theta}$; (ii) for every $\theta \geq \bar{\theta}$, one has*

$$(1.6) \quad V'_1(l_1^* + z_1(\theta)) = \theta V'_2(l_2^* + z_2(\theta)).$$

We shall not give a formal proof of this statement, which follows from elementary considerations. Note that in view of (1.3), the relation (1.6) may take the equivalent form

$$(1.7) \quad -z_1(\theta) V'_1(l_1^* + z_1(\theta)) = z_2(\theta) V'_2(l_2^* + z_2(\theta)) \quad \text{for every } \theta \geq \bar{\theta}.$$

The above lemma implies, in view of (1.2), that a consumer has a positive demand for money if and only if $\theta > \bar{\theta}$. We shall be concerned in the sequel with the case where the agents have a positive demand for money when the price of

the good is constant over time. This corresponds to the *Samuelson case*, according to the terminology of David Gale, in his excellent paper [30] on the overlapping generations model. Formally, we have the following assumption.

ASSUMPTION (1.d): $\bar{\theta} = V'_1(l_1^*)/V'_2(l_2^*)$ is less than 1.

The next fact gives more information about the variation of z_1 and z_2 with θ .

LEMMA 1.2: Assume (1.a) and (1.c). Then the restrictions of the excess demand functions $z_1(\theta)$ and $z_2(\theta)$ to the interval $[\bar{\theta}, +\infty)$ are continuously differentiable. For every $\theta \geq \bar{\theta}$,

$$\begin{aligned} z'_1(\theta) &= [V'_2(l_2^* + z_2(\theta)) + z_2(\theta) V''_2(l_2^* + z_2(\theta))]/\Delta, \\ z'_2(\theta) &= -[V'_1(l_1^* + z_1(\theta)) + z_1(\theta) V''_1(l_1^* + z_1(\theta))]/\Delta, \end{aligned}$$

in which $\Delta = V''_1(l_1^* + z_1(\theta)) + \theta^2 V''_2(l_2^* + z_2(\theta)) < 0$. In particular $z'_2(\theta) > 0$ for every $\theta \geq \bar{\theta}$. Moreover $z_2(\theta)$ diverges to $+\infty$ whenever θ tends to $+\infty$.

The first part of this statement is obvious by differentiating (1.6) or (1.7) and by using (1.3). On the other hand, the fact that $z_2(\theta)$ diverges to $+\infty$ whenever θ increases without bound is not difficult to verify. Indeed, we get from (1.3) that when θ tends to $+\infty$, $z_1(\theta)$ must tend to 0 if $z_2(\theta)$ remains bounded. But in that case, the left hand side of (1.6) is bounded while the right hand side diverges to $+\infty$. This is a contradiction which shows that $\lim_{\theta \rightarrow +\infty} z_2(\theta) = +\infty$.

The foregoing analysis gives some insight about the consequences upon a trader's behavior of a variation of θ (or of his "expected real interest rate" which is given by $\theta - 1$). A change of θ generates an *intertemporal substitution* as well as *income or wealth effects*. A rise of θ induces always an increase of $z_2(\theta)$ because intertemporal substitution and income effects work in the same direction for future consumption. On the other hand, the induced variation of $z_1(\theta)$ is ambiguous. We know indeed from Lemma 1.2 that $-z'_2(\bar{\theta})/z'_1(\bar{\theta}) = \bar{\theta}$. Therefore $z'_1(\bar{\theta}) < 0$, and thus by continuity $z'_1(\theta) < 0$ if θ is larger than but close enough to $\bar{\theta}$. However, the sign of $z'_1(\theta)$ is a priori indeterminate for large values of θ , because income and substitution effects are working in opposite directions for current consumption. As a matter of fact *the origin of the business cycles that are going to be analyzed in the present paper is precisely this conflict between intertemporal substitution and wealth effects*.

Examination of the expressions of $z'_1(\theta)$ and $z'_2(\theta)$ suggests that an important role in this regard should be played by the so-called Arrow-Pratt *relative degrees of risk aversion*

$$R_\tau(a_\tau) = -V''_\tau(a_\tau)/V'_\tau(a_\tau)$$

which are well defined whenever $a_\tau > 0$. These expressions measure in effect the degree of concavity (curvature) of each V_τ . We shall use the following assumption:

ASSUMPTION (1.e): $R_2(a_2)$ is a nondecreasing function of a_2 for every $a_2 > 0$.

For a justification of such an assumption (in a context involving uncertainty), see Arrow [1, Ch. 3]. It is then easy to get the following fact.

LEMMA 1.3: Assume (1.a) and (1.c). Then for every $\theta \geq \bar{\theta}$, $z'_1(\theta) < 0$ if and only if $R_2(l_2^* + z_2(\theta)) < (l_2^* + z_2(\theta))/z_2(\theta)$.⁷ Accordingly, (i) if $R_2(a_2) \leq 1$ for all $a_2 > 0$, then $z'_1(\theta) < 0$ for every $\theta \geq \bar{\theta}$; (ii) if (1.e) holds and if there exists $a_2 > 0$ such that $R_2(a_2) > 1$, then there exists a unique $\theta^* > \bar{\theta}$ such that $z'_1(\theta) < 0$ for every $\bar{\theta} \leq \theta < \theta^*$, $z'_1(\theta^*) = 0$ and $z'_1(\theta) > 0$ for every $\theta > \theta^*$.

The claim that $z'_1(\theta) < 0$ if and only if

$$R_2(l_2^* + z_2(\theta)) < (l_2^* + z_2(\theta))/z_2(\theta)$$

can be verified immediately by looking at the expression of $z'_1(\theta)$ in Lemma 1.2. Then if $R_2(a_2) \leq 1$ for all $a_2 > 0$, the left-hand side of this inequality never exceeds 1, while the right-hand side is always greater than 1 whenever $\theta \geq \bar{\theta}$, which shows (i). If Assumption (1.e) holds, the left-hand side of this inequality is a nondecreasing function of θ . When θ tends to $+\infty$, $z_2(\theta)$ tends also to $+\infty$ and thus under the assumptions of (ii), $R_2(l_2^* + z_2(\theta))$ exceeds 1 for θ large enough. On the other hand, $(l_2^* + z_2(\theta))/z_2(\theta)$ decreases from $+\infty$ to 1 when θ rises from $\bar{\theta}$ to $+\infty$. There is thus, by continuity, a unique $\theta^* > \bar{\theta}$ such that

$$R_2(l_2^* + z_2(\theta^*)) = (l_2^* + z_2(\theta^*))/z_2(\theta^*)$$

and it is clear that $z'_1(\theta) < 0$ whenever $\bar{\theta} \leq \theta < \theta^*$, $z'_1(\theta^*) = 0$ and $z'_1(\theta) > 0$ when $\theta > \theta^*$.

It may be useful to illustrate our findings by drawing in the plane (a_1, a_2) a consumer's offer curve, that is, the locus of all points of coordinates $a_1 = l_1^* + z_1(\theta)$, $a_2 = l_2^* + z_2(\theta)$ when θ varies. The result is shown in Figure 1. In view of (1.3), finding the point of the offer curve that corresponds to θ amounts to looking at its intersection with the intertemporal budget line of equation $\theta(a_1 - l_1^*) + (a_2 - l_2^*) = 0$. According to the previous lemmas, the offer curve is smooth and goes through the endowment point $A = (l_1^*, l_2^*)$ —this corresponds to $\theta \leq \bar{\theta}$. Its normal there is the vector $(\bar{\theta}, 1)$. The curve lies below the 45° line AB when $\bar{\theta} < \theta < 1$, and above when $\theta > 1$. Figures 1.a and 1.b are drawn under the assumption that $\bar{\theta} < 1$. Of course, if $\bar{\theta} \geq 1$, the curve would lie entirely above AB . Figure 1.a corresponds to the case considered in (i) of Lemma 1.3, in which $R_2(a_2) \leq 1$ for all $a_2 > 1$. The curve is then “monotone”, i.e. it has no critical point. Figure 1.b corresponds to the case in which (1.e) holds and in which $R_2(a_2) > 1$ for some $a_2 > 0$. The offer curve has then a unique critical point corresponding to the value $\theta = \theta^*$. When $\theta < \theta^*$ the substitution effect dominates, while the income effect is prominent for $\theta > \theta^*$.⁸

⁷ We allow the value $+\infty$ when $\theta = \bar{\theta}$.

⁸ W. Brock brought to my attention after the completion of this work that an earlier paper by Brock and Scheinkman [11] contains a discussion of the slope of a trader's offer curve in relationship to the coefficients of relative risk aversion.

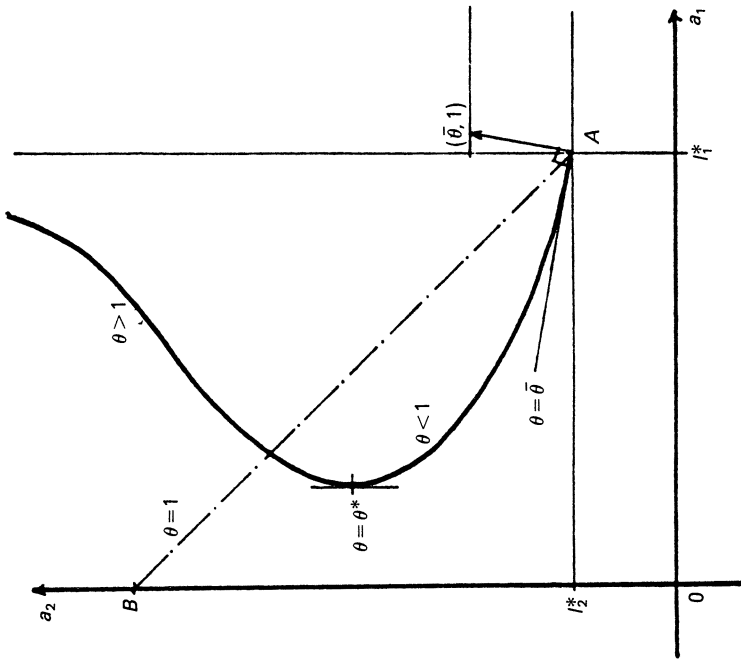


FIGURE 1.b

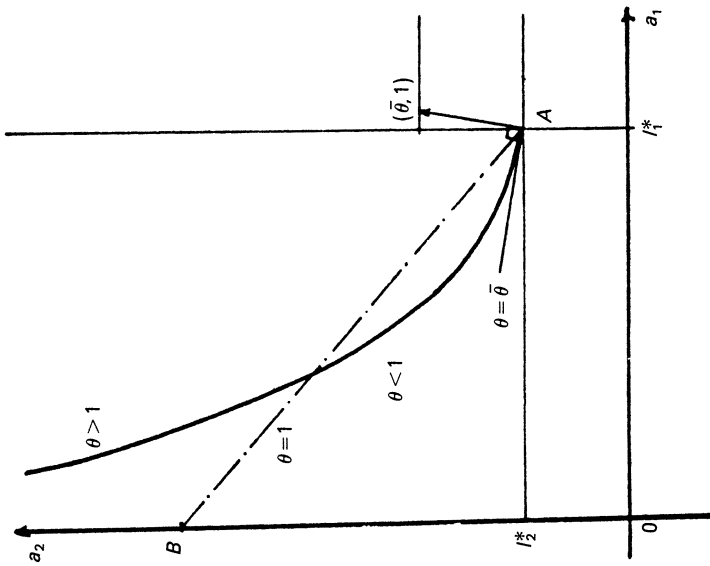


FIGURE 1.a

2. PERIODIC COMPETITIVE EQUILIBRIA

We now turn attention to the definition of periodic equilibria with perfect foresight.

The evolution of the economy is described by a sequence of temporary competitive equilibria. Since an old trader's excess demand for the good at any date is equal to the real value of his money stock, equilibrium at date t for the good and for money is given by

$$(2.1) \quad z_1(p_t/p_{t+1}^e) + M/p_t = 0,$$

$$(2.2) \quad m^d(p_t, p_{t+1}^e) = M,$$

in which p_t is the current price and p_{t+1}^e the price that is expected to prevail at the next date.⁹ The two equations are of course equivalent by Walras's Law; see (1.2).

A sequence of competitive equilibria with *perfect foresight* is a sequence (p_t) that is a solution of (2.1) or (2.2) with $p_{t+1}^e = p_{t+1}$ for all t . It is a straightforward matter to see that under Assumptions (1.a), (1.c), such a sequence is a solution of the following system (and conversely):

$$(2.3) \quad z_1(\theta_t) + z_2(\theta_{t-1}) = 0,$$

$$(2.4) \quad p_{t+1}z_2(\theta_t) = M,$$

in which $\theta_t = p_t/p_{t+1} > \bar{\theta}$ for all t . Indeed, in view of (1.2), (2.4) is nothing else than (2.2) with $p_{t+1}^e = p_{t+1}$. This relation tells us that an old trader's excess demand for the good at an arbitrary date t , i.e. M/p_t , is under perfect foresight what he planned to do when he was young, i.e. $z_2(\theta_{t-1})$. Replacing M/p_t by $z_2(\theta_{t-1})$ in (2.1), with $p_{t+1}^e = p_{t+1}$, yields (2.3). The fact that M is positive implies $z_2(\theta_t) > 0$ and thus $\theta_t > \bar{\theta}$.

The system (2.3), (2.4) describes the dynamics of the economy under perfect foresight.¹⁰ The important property of the system is that it *dichotomizes*. The good market equation (2.3) determines the equilibrium sequence of θ_t —or equivalently of real interest rates $\theta_t - 1$ —and indeed all real equilibrium quantities, independently of (2.4), i.e. of the level of the money stock M . The level of equilibrium prices is in turn determined by the money equation, i.e. through $p_{t+1} = M/z_2(\theta_t)$.¹¹ In fact equilibrium prices are, under perfect foresight, proportional to the money stock M . This is the traditional Quantity Theory (*neutrality of money*).

A *periodic (monetary) equilibrium with perfect foresight* is defined as a periodic sequence (p_t) that satisfies (2.1) and (2.2) with $p_{t+1}^e = p_{t+1}$ for all t , alternatively (2.3) and (2.4). From the above discussion, such a periodic sequence (p_t) can

⁹ Equilibrium on the labor market has already been taken into account by setting the real wage equal to one and by assuming that output is equal to the traders' labor supply.

¹⁰ That sort of system has been used by numerous writers when studying the dynamics of the overlapping generation model with perfect foresight. See e.g. Azariadis [2], Cass [12], Benhabib and Day [7], Gale [30], Tirole [57]. The same sort of approach has been used in Grandmont [33] to study monetary steady states in a model involving several goods.

¹¹ One verifies easily, by using (2.3) and (1.3), that $p_t/p_{t+1} = z_2(\theta_t)/z_2(\theta_{t-1}) = -z_2(\theta_t)/z_1(\theta_t) = \theta_t$.

be identified with a periodic sequence (θ_t) that fulfills (2.3) with $\theta_t > \bar{\theta}$ for all t , through the relation $p_{t+1} = M/z_2(\theta_t)$. Such a periodic sequence (p_t) or (θ_t) , say with period k , may be described by the k consecutive values that it takes, i.e. with its *periodic orbit* $(p_1^*, \dots, p_k^*)$ or $(\theta_1^*, \dots, \theta_k^*)$.

The system (2.3), (2.4) induces a well defined (but fictitious) *backward perfect foresight (b.p.f.) dynamics*. Under Assumptions (1.a), (1.c), the restriction of z_2 to the interval $[\bar{\theta}, +\infty)$ is increasing and maps that interval onto $[0, +\infty)$ (Lemma 1.2). It has therefore an inverse. Then (2.3) and (2.4) give rise to very simple difference equations of the form

$$(2.5) \quad \theta_{t-1} = z_2^{-1}(-z_1(\theta_t)) \equiv \varphi(\theta_t),$$

$$(2.6) \quad p_t = p_{t+1} z_2^{-1}(M/p_{t+1}) \equiv \Phi(p_{t+1}).$$

The b.p.f. map φ on real interest rates is continuously differentiable and maps $[\bar{\theta}, +\infty)$ into $[\bar{\theta}, \bar{\theta})$ with $\bar{\theta} = z_2^{-1}(l_1^*)$. Then a periodic equilibrium with perfect foresight $(\theta_1^*, \dots, \theta_k^*)$ can be identified to a *cycle* $(\theta_k^*, \dots, \theta_1^*)$ —in which time has been reversed—of the b.p.f. map φ such that $\theta_i^* > \bar{\theta}$ for all i .¹² Similarly the b.p.f. map Φ on *equilibrium prices* takes $(0, +\infty)$ into itself and a periodic equilibrium with perfect foresight may be identified to a cycle $(p_k^*, \dots, p_1^*)$ of Φ . Of course the b.p.f. dynamics associated to φ and Φ are the same since one gets one map from the other through the change of variable $p_{t+1} = M/z_2(\theta_t)$.

Figure 2, which is drawn under the assumption $\bar{\theta} < 1$, illustrates the b.p.f. dynamics associated with (2.5) and the map φ by using a trader's offer curve. Starting with $\theta_t > \bar{\theta}$, we first draw the intertemporal budget line of equation $\theta_t a_1 + a_2 = \theta_t l_1^* + l_2^*$. It intersects the trader's offer curve at the point of coordinates $(l_1^* + z_1(\theta_t), l_2^* + z_2(\theta_t))$. Finding the value of θ_{t-1} that satisfies $z_1(\theta_t) + z_2(\theta_{t-1}) = 0$ is then achieved by following the arrows on the Figure, by going first vertically to the 45° line AB that passes through the endowment point (l_1^*, l_2^*) and then horizontally back to the offer curve. This procedure yields the point of coordinates $(l_1^* + z_1(\theta_{t-1}), l_2^* + z_2(\theta_{t-1}))$ and thus the corresponding value of $\theta_{t-1} = \varphi(\theta_t)$ by drawing the corresponding intertemporal budget line. The Figure shows that φ has always a fixed point $\theta = \bar{\theta}$, which corresponds to a nonmonetary stationary state, and that $\theta = 1$ is the unique monetary stationary state whenever $\bar{\theta} < 1$. The Figure gives moreover an example of a cycle of period 2.

3. STABILITY AND LEARNING

This Section is concerned with the stability of periodic competitive equilibria with perfect foresight. We look first at the stability of such equilibria under the dynamics with perfect foresight that are generated by the equations (2.3) and

¹² A cycle of a map is most conveniently defined through its iterates. The *iterates* of a function f of a set X into itself are defined recursively by $f^1(x) = f(x)$, $f^i(x) = f(f^{i-1}(x))$. A *periodic orbit* (or a *cycle*) of (primitive) period k of the map f is then defined by $(x_1^*, \dots, x_k^*)$ such that (i) x_1^* is a fixed point of f^k , i.e. $x_1^* = f^k(x_1^*)$, and (ii) $x_i^* = f^{i-1}(x_1^*) \neq x_1^*$ for all $i = 2, \dots, k$. This definition implies of course that all x_i^* of the orbit are fixed points of f^k , and that they all differ. In particular $(x_1^*, \dots, x_k^*, x_1^*, \dots, x_{k-1}^*)$ defines the same periodic orbit.

Perfect Foresight Dynamics

We assume throughout (1.a) and (1.c). Consider a periodic competitive equilibrium with perfect foresight (p_t), with period k , or equivalently its periodic orbit $(p_1^*, \dots, p_k^*)$. We wish to assess the stability properties of such an equilibrium under the perfect foresight dynamics associated with the equations (2.3), (2.4).

The most natural step is to answer this question first by looking at the backward perfect foresight dynamics. Indeed $(p_k^*, \dots, p_1^*)$ is a cycle of period k of the b.p.f. map Φ . Since this means in particular that p_k^* is a fixed point of Φ^k , we may say that the cycle is (locally) *stable in the b.p.f. dynamics* if and only if there exists a neighborhood I of p_k^* such that for every p in I , the orbit of p by the map Φ^k stays in I and converges to p_k^* , i.e. $\Phi^{kj}(p) \in I$ for all $j \geq 1$ and $\lim_{j \rightarrow +\infty} \Phi^{kj}(p) = p_k^*$ (by continuity, this ensures that $\Phi^{kj}(\Phi^i(p))$ converges to p_{k-i}^* as well for $i = 1, \dots, k-1$). Since Φ is continuously differentiable, this condition is equivalent to stating that *the absolute value of the derivative of Φ^k at p_k^* , i.e. $|D\Phi^k(p_k^*)|$, is less than 1*. In order to make sense, of course, this definition should not depend upon the point chosen on the periodic orbit, since $(p_1^*, \dots, p_1^*, p_k^*, \dots, p_{k-1}^*)$ describes the same cycle. In fact we get, by employing the chain rule of differentiation,

$$\begin{aligned} (3.1) \quad D\Phi^k(p_k^*) &= D\Phi^{k-1}(p_{k-1}^*)D\Phi(p_k^*) \\ &= D\Phi(p_1^*) \cdots D\Phi(p_k^*) \\ &= D\Phi^k(p_1^*). \end{aligned}$$

The definition is thus consistent.¹⁵

Yet, this notion of stability is not entirely satisfactory since after all, time should go *forward* in any meaningful dynamics. The problem is however that a forward perfect foresight dynamics is not well defined by (2.3), (2.4). This is most easily seen by looking at Figure 2. If one starts from the point of the offer curve that corresponds to θ_{t-1} , the forward dynamics associated with (2.3) require that one should go first horizontally to the 45° line AB and then vertically back to the offer curve. In the case of Figure 2 this procedure yields two points. If one goes back to Figure 1.a, one sees easily that the forward dynamics implied by (2.3) is eventually undefined in that case, if one starts with a value of θ that is greater than 1. The reason is that at some stage one should have $z_2(\theta_{t-1}) > l_1^*$, and thus there could not exist a θ_t such that $z_1(\theta_t) + z_2(\theta_{t-1}) = 0$.

A *forward perfect foresight dynamics* can be defined nevertheless *locally* in a neighborhood of a periodic orbit $(p_1^*, \dots, p_k^*)$ provided that $D\Phi^k(p_i^*) \neq 0$. Indeed this implies in view of (3.1) that $D\Phi(p_i^*) \neq 0$, so that Φ is locally invertible around each element p_i^* of the periodic orbit. Then if the cycle is unstable (resp.

¹⁵ It is clear that we would have arrived at an equivalent definition by considering instead the b.p.f. dynamics on real interest rates generated by (2.3) and the b.p.f. map φ .

stable) in the backward perfect foresight dynamics, it will be stable (resp. unstable) in this local forward perfect foresight dynamics.¹⁶

Stability and Learning

It should be clear that perfect foresight is very unlikely to obtain out of “long run” (here periodic) equilibria. At any date, a trader has no way to decide which price will prevail in the future on the sole basis of observed current and past prices, unless his environment has been repetitive enough in the past. These remarks lead to the conclusion that one should portray the traders as learning the dynamics of prices on the transition path toward periodic equilibria. After all, the usual justification of the mere assumption of self-fulfilling expectations is that learning has been “completed.” Taking due account of learning when formulating the dynamics of the economy yields a decentralized theory of expectations formation—which the postulate of self-fulfilling expectations falls short of achieving—and may have important consequences, as we shall see, on the stability properties of periodic equilibria.

One way to model expectations formation is to postulate that at each date, the price p_{t+1}^e that is expected by the young agent is a fixed function of current and past prices

$$(3.2) \quad p_{t+1}^e = \psi(p_t, p_{t-1}, \dots, p_{t-T})$$

in which the memory lag T is assumed to be “large” and independent of time.¹⁷ A *temporary competitive equilibrium* (with learning) at t is then a price p_t that satisfies the equilibrium conditions (2.1), (2.2) with p_{t+1}^e being given by (3.2). This yields

$$(3.3) \quad z_1(p_t / \psi(p_t, \dots, p_{t-T})) + (M/p_t) = 0,$$

$$(3.4) \quad m^d(p_t, \psi(p_t, \dots, p_{t-T})) = M.$$

The next assumption on the expectation function is made to guarantee the existence of a solution p_t to the above system of equations, when past prices are given. It is there essentially to ensure enough intertemporal substitution when the current price p_t varies.

ASSUMPTION (3.f): *The expectation function ψ is continuously differentiable. The elasticity of ψ with respect to the current price is between 0 and 1, that is¹⁸*

$$0 \leq \psi'_0(p_t, \dots, p_{t-T}) p_t / \psi(p_t, \dots, p_{t-T}) \leq 1.$$

¹⁶ An approximation to the local forward perfect foresight dynamics is obtained by linearizing the b.p.f. map Φ around each p_t^* , and by inverting the linearized dynamics. An equivalent local forward perfect foresight dynamics on equilibrium real interest rates would be obtained, of course, by inverting locally the map φ .

¹⁷ The analysis that follows would be unchanged if a trader were informed of past quantities as well. The analysis would have been also unaltered had we assumed differing expectations functions among the members of a generation.

¹⁸ The notation $\psi'_j(p_t, \dots, p_{t-T})$ stands for the partial derivative of ψ with respect to p_{t-j} for $j = 0, 1, \dots, T$.

Moreover, the ratio $p_t/\psi(p_t, \dots, p_{t-T})$ exceeds eventually $\bar{\theta}$ when θ goes to $+\infty$.

We may then state the following proposition.

PROPOSITION 3.1: *Assume (1.a) and (1.c). Then under Assumption (3.f), there exists a unique temporary competitive equilibrium price p_t , given the past prices $p_{t-1}, \dots, p_{t-T}$. The temporary Walrasian equilibrium function so defined $p_t = W(p_{t-1}, \dots, p_{t-T})$ is continuously differentiable.*

The proof of this claim is simple. By virtue of Walras's Law, we may focus attention on the good market equation (3.3) alone. Then when p_t goes to 0, the real balance M/p_t goes to infinity while the young consumer's excess demand for the good is bounded below by $-l_1^*$. Thus we are sure that excess demand goes to $+\infty$ on the good market when p_t decreases to 0. This is due to the real balance effect, which operates on the old trader alone.

We wish now to show that an excess supply of the good appears when its price increases without bound. Or equivalently—again by Walras's Law—that an excess demand for money appears in such a circumstance. But that is easy if one considers the money equation (3.4) which can be put in the following equivalent but more convenient form by using (1.2):

$$(3.5) \quad \psi(p_t, \dots, p_{t-T})z_2(p_t/\psi(p_t, \dots, p_{t-T})) = M.$$

Under Assumption (3.f), ψ and p_t/ψ are nondecreasing functions of p_t . Thus the left hand side of (3.5) is nondecreasing and must go to $+\infty$ when p_t tends to $+\infty$. For otherwise, one could find a particular sequence p_t^n that tends to $+\infty$ such that both $\psi(p_t^n, \dots, p_{t-T})$ and $p_t^n/\psi(p_t^n, \dots, p_{t-T})$ are bounded, a contradiction. The young trader's demand for money exceeds M when p_t is large enough. Here it is the intertemporal substitution effect that acts upon the young trader that plays a central role.¹⁹

By continuity (3.3) has a solution in p_t . In fact, it is easy to verify that under Assumption (3.f), the left-hand side of (3.5) is increasing whenever it is positive, which proves uniqueness. Continuous differentiability of the temporary Walrasian equilibrium function W is then a consequence of the implicit function theorem.²⁰

It may be noted in addition that *under the assumptions of the Proposition, the unique temporary equilibrium price p_t is stable in any Walrasian tatonnement process at date t in which prices respond positively to excess demand.*

Proposition 3.1 establishes that under Assumption (3.f), the *forward dynamics with learning* generated by (2.1), (2.2), and (3.2) is well defined and is described

¹⁹ For a further discussion of the respective roles of wealth and intertemporal substitution effects in a competitive monetary economy, see Grandmont [33].

²⁰ The fact that the equilibrium price p_t at t depends only on past prices is due to the simplified structure of the model. For instance, if there were several different consumers, the current equilibrium would depend also on the initial distribution of money holdings. If utilities were not separable over time, it would depend also on past consumptions. In general, current equilibrium prices would depend on past equilibrium states (prices, consumptions, money holdings, and so on). See, e.g., Grandmont and Hildenbrand [35], Grandmont and Laroque [36].

by the difference equation

$$(3.6) \quad p_t = W(p_{t-1}, \dots, p_{t-T}).$$

It should be clear that any solution (p_t) of (3.6) that has period k is a periodic competitive equilibrium with perfect foresight (is a solution of (2.3), (2.4)) with the same period, and conversely, provided that traders are able to “recognize” periodicity k , or more precisely, if they satisfy the following assumption.

ASSUMPTION (3.g): *The expectation function ψ is consistent with periodicity k . For any sequence $p_t, p_{t-1}, \dots, p_{t-T}$ that displays period k , that is, such that $p_{t-j} = p_{t-j-k}$ for $j = 0, 1, \dots$,*

$$\psi(p_t, \dots, p_{t-T}) = p_{t-k+1}.$$

Indeed, this assumption implies that agents “learn” and thus may make mistakes when their environment is chaotic, but that they make correct forecasts along any solution of (3.6) that has period k , which proves the claim.

We wish now to establish some relation between the stability of a given cycle with period k under the backward perfect foresight dynamics (Φ or φ -stability) and its stability properties under the forward dynamics with learning implied by (3.6) (W -stability). It may seem foolish at first sight to hope to ascertain such a relationship. After all, the b.p.f. map Φ describes a fictitious and very abstract dynamics that goes backward, while the dynamics with learning go “forward.” The enterprise is not as vain as one might expect, however. Under Assumption (3.g), anticipations are correct along the periodic orbit, and thus by continuity, nearly correct in its neighborhood. On the other hand, the essence of learning is precisely to forecast future prices by looking backward at past prices.

Before proceeding, we must make precise what we mean by a cycle that is stable in the dynamics with learning. Assume accordingly (1.a), (1.c), (3.f) and consider a periodic solution (p_t) of (3.6) with period k , or equivalently its periodic orbit $(p_1^*, \dots, p_k^*)$. One may in fact look at (3.6) as defining a dynamic system in a space of larger dimension. For any $q_{t-1} = (p_{t-1}, \dots, p_{t-T})$, let us consider $q_t = \tilde{W}(q_{t-1})$ that is determined by

$$(3.7) \quad q_t = (W(p_{t-1}, \dots, p_{t-T}), p_{t-1}, \dots, p_{t-T+1}).$$

This procedure yields a map $\tilde{W}$ that takes the interior of $\mathbb{R}_+^T$ into itself and that generates the same dynamics as W . In particular, if one defines for every $i = 1, \dots, k$, the following vector of $\mathbb{R}^T$,

$$(3.8) \quad q_i^* = (p_i^*, \dots, p_1^*, p_k^*, \dots, p_1^*, \dots),$$

then $(q_1^*, \dots, q_k^*)$ is a cycle of $\tilde{W}$, and conversely.

We may say therefore that *the cycle $(p_1^*, \dots, p_k^*)$ is (locally) stable under the dynamics (3.6) with learning if and only if the associated cycle $(q_1^*, \dots, q_k^*)$ is a stable cycle of the map $\tilde{W}$; or equivalently, since q_1^* is a fixed point of $\tilde{W}^k$, if and*

only if the eigenvalues of the Jacobian matrix of $\tilde{W}^k$ at q_1^* , i.e. $D\tilde{W}^k(q_1^*)$, have all a modulus less than 1.

In order to make sense, here again, the definition should not depend upon whether we deal with q_1^* or any other point q_i^* of the periodic orbit $(q_1^*, \dots, q_k^*)$ of $\tilde{W}$. From $\tilde{W}^k(q) = \tilde{W}^{k-1}(\tilde{W}(q))$, we get by employing the chain rule of differentiation,

$$(3.9) \quad D\tilde{W}^k(q_1^*) = D\tilde{W}^{k-1}(q_2^*)D\tilde{W}(q_1^*) = \dots = D\tilde{W}(q_k) \dots D\tilde{W}(q_1^*).$$

Thus $D\tilde{W}^k(q_1^*)$ is obtained as the product of the Jacobian matrices of $\tilde{W}$, $D\tilde{W}(q_i^*)$, at the different points of the cycle. Applying the same procedure to $D\tilde{W}^k(q_i^*)$, we obtain

$$D\tilde{W}^k(q_i^*) = D\tilde{W}(q_{i-1}^*) \dots D\tilde{W}(q_1^*)D\tilde{W}(q_k^*) \dots D\tilde{W}(q_i^*).$$

Now for any two square matrices A and B , the eigenvalues of the products AB and BA are the same (see e.g. Wilkinson [51, p. 54]), so that the eigenvalues of the Jacobian matrices $D\tilde{W}^k(q_1^*), \dots, D\tilde{W}^k(q_k^*)$, are all the same, which establishes the equivalence we were looking for.

We are able now to state that a cycle of period k that is stable in the *backward* perfect foresight dynamics, is stable in the *forward* dynamics with learning, whenever a trader facing a periodic sequence of prices $(p_t, \dots, p_{t-T})$ with period k makes essentially a forecast that is (locally) an average of the “relevant” past prices p_{t-k+1} , p_{t-2k+1} , and so on. Formally, consider the following assumption.

ASSUMPTION (3.h): For any periodic sequence $(p_1, \dots, p_{t-T})$ that has period k , the expectation function satisfies $|\psi'_0| \leq \varepsilon$ and $|\psi'_j| \leq \varepsilon$ for all $j \geq 1$ that are not of the form $j = nk - 1$ for some integer $n \geq 1$. All other derivatives ψ'_j with $j = nk - 1 \geq 1$ are nonnegative.

It is instructive to see the implications of (3.h) in the light of Assumption (3.g), in the limiting case where $\varepsilon = 0$. Consider a sequence of prices that has period k $(\bar{p}_t, \dots, \bar{p}_{t-T})$. Then for any (even aperiodic) sequence $(p_t, \dots, p_{t-T})$ in its neighborhood, a trader's expected price is approximately given by

$$\psi(p_t, \dots, p_{t-T}) \approx \psi(\bar{p}_t, \dots, \bar{p}_{t-T}) + \sum_{j=0}^T (p_{t-j} - \bar{p}_{t-j}) \psi'_j(\bar{p}_t, \dots, \bar{p}_{t-T})$$

or

$$\psi(p_t, \dots, p_{t-T}) \approx \sum_{j=0}^T p_{t-j} \psi'_j(\bar{p}_t, \dots, \bar{p}_{t-T})$$

since $\psi(\lambda \bar{p}_t, \dots, \lambda \bar{p}_{t-T})$ is homogeneous of degree 1 with respect to λ under Assumption (3.g). On the other hand, this assumption implies that the expected price ψ is unchanged for each integer $0 \leq j \leq k-2$ whenever the constant sequence $(\bar{p}_{t-j}, \bar{p}_{t-j-k}, \dots, \bar{p}_{t-j-nk}, \dots)$ is multiplied by an arbitrary real number $\lambda > 0$, and that ψ changes proportionately to λ when $j = k-1$. One gets thus by differentiation

with respect to λ in each case

$$\sum_{n \geq 1} \psi'_{j+(n-1)k} = 0 \quad \text{for} \quad 0 \leq j \leq k-2 \quad \text{and} \quad \sum_{n \geq 1} \psi'_{nk-1} = 1$$

where the derivatives are evaluated at $(\bar{p}_t, \dots, \bar{p}_{t-T})$. Then condition (3.h) with $\varepsilon = 0$ implies

$$\psi(p_t, \dots, p_{t-T}) = \sum_{n \geq 1} p_{t-nk+1} \psi'_{nk-1}(\bar{p}_t, \dots, \bar{p}_{t-T}),$$

the first term of this sum being 0 when $k = 1$. The trader's forecast is locally some sort of "average," with nonnegative weights, of the relevant past prices p_{t-k+1} , p_{t-2k+1} , and so on (the current price being given no weight when $k = 1$). If ε is "small," the trader's behavior is close to the one we just described, which does not sound implausible. Then we have the following proposition:²¹

PROPOSITION 3.2: *Assume (1.a), (1.c), and (3.f). Under Assumption (3.g), a cycle $(p_1^*, \dots, p_k^*)$ with perfect foresight with period k is a periodic solution of the dynamics with learning (3.6) with the same period and vice versa. Under the additional Assumption (3.h), stability of $(p_k^*, \dots, p_1^*)$ under the backward perfect foresight dynamics (Φ or φ -stability) implies stability in the forward dynamics with learning (W -stability) provided that ε is small enough.*

Assumptions (3.g), (3.h) pertain to a *given* periodicity k . A possible interpretation of Proposition 3.2 is then that the model has been previously in a periodic equilibrium with period k and that it is slightly perturbed. It is then plausible that the agents, having previously "recognized" the periodicity—through some unspecified process—use an expectation function satisfying Assumptions (3.g) and (3.h) on the transition path in the perturbed economy.

We would like however to have a more *global* story. The ideal situation would be to have an expectation function that is consistent, in the sense of Assumption (3.g), with *all* periods, which is clearly impossible in a finite memory T . Nevertheless, given any finite set of periodicities $K = \{k_1, \dots, k_s\}$, there are expectation functions that are consistent with all periods in K , provided that T is greater than the *least common multiple* k^* of all periods k_i in K : it suffices to take for all sequences $(p_t, \dots, p_{t-T})$ that display any periodicity k_i in K , $\psi(p_t, \dots, p_{t-T}) = p_{t-k^*+1}$. This may impose that the memory lag be extraordinarily large if K is chosen arbitrarily—think of the case where $K = \{1, 2, 3, \dots, 10\}$. Presumably, however, the traders will tend to restrict the set K of periods that they wish to be able to recognize on the basis of their knowledge of the set of periodic equilibria with perfect foresight of the economy under consideration. A particular instance obtains in the present model because, as we shall see in the next section, periods that are equal to a power of 2 should "appear first." A plausible behavior of the

²¹ The proof of this claim, which is mostly computational, is given in the Appendix in order to ease the exposition.

agents would be then to choose a set of allowable periods of the form $\{1, 2, 4, \dots, 2^n\}$ with $2^n < T$.²²

The outcome of the foregoing analysis is that *one should generally be very cautious when interpreting the stability results that are obtained in a model employing the convenient but abstract assumption that traders have perfect foresight even out of “steady states”*—as has been customary recently under the impulse of the so-called “New Macroeconomics” school. *For if the traders’ learning processes are taken explicitly into account when modeling the dynamic evolution of an economy—as they should be since learning is after all the primitive concept that enables model builders to justify at least implicitly the mere assumption of perfect foresight—then stability results may well be reversed.* The point is most forcefully illustrated by the case represented in Figure 1.a. The stationary equilibrium $\theta = 1$ then is unstable in the forward perfect foresight dynamics (equivalently, the stationary price $p^* = M/z_2(1)$ is Φ -stable). Suppose now that traders use in fact a learning process to form their forecasts and that the associated expectation function ψ satisfies for every stationary sequence of prices ($p_t = p$ for all t) $\psi(p, p, \dots, p) = p$, and that it does not depend on the current price. Then ψ satisfies (3.g) and (3.h), with $k = 1$ and $\varepsilon = 0$ and Proposition 3.2 tells us that the stationary equilibrium, being stable in the b.p.f. dynamics, is indeed stable when traders employ the above learning process, which appears quite reasonable.

REMARK: The assumption that traders use the same expectation function ψ over time may be interpreted as follows. Consumers have a *given* “model” of the workings of the economy and use a *given* statistical technique (Bayesian methods or maximum likelihood procedures) to estimate the parameters of the model and to forecast future prices. This way of proceeding yields the map ψ .

Stability of a stationary monetary equilibrium has been studied in the more general context of temporary equilibrium theory by Fuchs and Laroque [29]; see also Tillman [56]. The issue of stability of a stationary equilibrium when traders employ more sophisticated learning procedures, in particular when they may revise their expectation functions over time in view of previous forecasting errors, has been investigated by Fuchs [24, 25, 26, 27, 28]. Stability of a *periodic*

²² The above discussion is intended only to be suggestive. It is clear that the choice of an expectation function and thus of the set K and the memory lag T , and more generally the learning behavior of the agents, should be modelled as the outcome of some optimizing behavior. The present analysis has nothing to say on that matter.

We already noted that a cycle with learning with a period in K is a cycle with perfect foresight. Conversely, a periodic equilibrium with perfect foresight with a period in K is a cycle of the dynamics with learning. It is perhaps worth emphasizing however that the dynamics with learning may have other cycles, with a period *not* in K , along which traders make incorrect forecasts. One might think that some of these problems would be alleviated if one set $T = +\infty$. This may not be so, however. Mike Woodford pointed out to me that even in that case there may be no expectation function having continuous partial derivatives and satisfying (3.g), (3.h) for all k , with $\varepsilon = 0$. The reason is that all partial derivative ψ'_j evaluated at constant sequences $(p, p, \dots)$ are then 0, which would seem to contradict $\psi(p, p, \dots) = p$ for all p . The difficulty is likely to remain if ε is “small.” The issue appears to me still open, however, as the foregoing argument is not mathematically tight.

competitive equilibrium when consumers revise their expectations functions over time, and its relation to stability in the backward perfect foresight dynamics, is an open issue. Note that the problem becomes much more complicated, since the state of the economy at a given date must then be described by a price *and* an expectation function. The paper by Fuchs and Laroque [29] contains also a few partial results concerning local dynamics in a neighborhood of a given cycle.

The question of the stability of perfect foresight or rational expectations equilibria when traders employ a given learning procedure has also been addressed in a macroeconomic stochastic model by M. Bray [10], P. Champsaur [15], and B. Friedman [22].

4. EXISTENCE, STABILITY, AND BIFURCATION OF PERIODIC EQUILIBRIA

The results that were obtained in the preceding two Sections imply that finding a periodic monetary equilibrium with perfect foresight is equivalent to finding a periodic orbit of the b.p.f. map $\theta_{t-1} = \varphi(\theta_t)$ with $\theta_t > \bar{\theta}$ for all t . We study in the present Section the conditions that lead to the occurrence of cycles of the map φ , their multiplicity and stability properties. We concentrate on the b.p.f. map φ , rather than Φ , because its properties can be related most easily to the shape of the traders' offer curve, as shown in Figure 2. The tools that we shall use are borrowed from recent mathematical theories that have been constructed by using partly the notion of the “*bifurcation*” of a dynamical system in order to explain the emergence of cycles and the transition to turbulent (“chaotic”) behavior in physical, biological, or ecological systems.²³

The general idea is that cycles with a period $k \geq 2$ will appear in the present model whenever there is an important conflict between the intertemporal substitution effect and the wealth effect that result from a variation of the real interest rate, so that a trader's offer curve displays a significant “hump” as in Figure 2. For this to be the case, it is necessary that the concavity of an old trader's utility function as measured by $R_2(a_2)$ be greater than 1 for some $a_2 > 0$. The phenomenon will be seen to appear in particular, in the constant elasticity case, when old consumers have utility functions that are sufficiently more concave than those of the young traders.

A most interesting feature of this model is the coexistence of cycles with different periods. An instance of the phenomenon was already given in Figure 2, in which a cycle with period 2 coexisted with the unique monetary steady state. Much more can be said in fact. If the set of positive integers is ordered in a specific way, which we may call the “Sarkovskii's ordering” from the name of the mathematician who discovered it, and if the map φ has a cycle of period k ,

²³ One important mathematical reference in this field is Collet and Eckmann [16]. For an excellent review of various applications of the theory, see May [48]. Part of this theory has been applied already in economics or game theory, in particular Benhabib and Day [6, 7], Dana and Malgrange [17], Day [18, 19], Jensen and Urban [39], Rand [49]. Some parts of this theory that seemed (to me) relevant and useable by economic theorists are reviewed in Grandmont [34].

then we are sure that it has also a cycle with a period k' in which k' is *any* integer that is ranked before k in the ordering. In particular, if there is a cycle of period 2^k , then there is also periodic orbit with period 2^n in which $n = 0, 1, \dots, k-1$. Moreover, if there is a cycle of period 3, then cycles of period n , in which n is an *arbitrary* positive integer, will also exist.

In the face of this bewildering multiplicity of periodic equilibria, the stability issue is essential. There is indeed a condition on the traders' offer curve, namely that it has a negative "Schwarzian derivative," that ensures that there exists *at most* one periodic orbit that is stable in the b.p.f. dynamics. This condition will be related to certain properties of the traders' utility functions. It will be seen in particular that the condition obtains, in the constant elasticity case, when young consumers are only moderately risk averse (have a relative risk aversion that is less than or equal to 1) while old traders have a relative risk aversion greater than or equal to 2. In such circumstances, *all* other cycles of φ are unstable in the b.p.f. dynamics and are thus stable in the local forward perfect foresight dynamics, as defined in Section 3. If the traders' expectation function ψ is consistent with the periodicity of the unique φ -stable cycle, whenever it exists, and if ψ satisfies (3.h), then the cycle is also stable in the dynamics with learning (Proposition 3.2). Needless to say, this does *not* imply in general that the dynamics with learning have a unique stable cycle.

The last part of the Section will be devoted to the analysis of the "bifurcations" of the dynamical system, and more precisely of the emergence of φ -stable cycles. It will be seen essentially that if one considers a one parameter family of economies, then periodic orbits that are stable in the b.p.f. dynamics do appear once again in conformity to the Sarkovskii's ordering of the set of integers. We shall give a particular example of such a bifurcation scheme in the case of a constant relative risk aversion, in which the young consumers' risk aversion is held fixed, while the old traders' risk aversion is increased progressively.

Characteristics of the Map φ and Equivalent Dynamics

We begin the analysis by reviewing a few basic properties of the map φ and look at dynamical systems that are equivalent to the difference equation $\theta_{t-1} = \varphi(\theta_t)$. These equivalent dynamical systems are in fact obtained by making a change of variable $\tilde{\theta} = h(\theta)$.

We assume throughout this Section (1.a), (1.c). Then the function

$$(4.1) \quad \varphi(\theta) = z_2^{-1}(-z_1(\theta)) = z_2^{-1}(z_2(\theta)/\theta)$$

maps the interval $[\bar{\theta}, +\infty)$ into $[\bar{\theta}, \bar{\bar{\theta}}]$ in which $\bar{\bar{\theta}} = z_2^{-1}(I_1^*)$, and is continuously differentiable. Clearly $\bar{\theta}$ is a fixed point of φ , that corresponds to a nonmonetary stationary equilibrium. On the other hand, we have $\varphi(\theta) < \theta$ whenever $\theta > 1$. Thus if $\bar{\theta} \geq 1$, there can be no monetary cycle of any period since $\varphi^j(\theta)$ decreases monotonically to $\bar{\theta}$ as j tends to infinity. We shall assume accordingly $\bar{\theta} < 1$ (Assumption (1.d)). We have, then, Lemma 4.1.

LEMMA 4.1: Assume (1.a), (1.c), (1.d). Then φ maps the interval $[\bar{\theta}, +\infty)$ into $[\bar{\theta}, \bar{\theta})$ in which $\bar{\theta} = z_2^{-1}(l_1^*)$, and is continuously differentiable. Moreover (i) $\varphi(\bar{\theta}) = \bar{\theta}$, $\varphi(1) = 1$, $\varphi(\theta) > \theta$ whenever $\bar{\theta} < \theta < 1$ and $\varphi(\theta) < \theta$ whenever $\theta > 1$; (ii) one has $\varphi'(\bar{\theta}) = 1/\bar{\theta} > 1$ and $\varphi'(1) < 1$; (iii) let $\alpha_2 = \sup R_2(a_2)$; then if $\alpha_2 \leq 1$, one has $\varphi'(\theta) > 0$ for all $\theta \geq \bar{\theta}$; (iv) if $\alpha_2 > 1$ and if $R_2(a_2)$ is nondecreasing (assumption (1.e)), then the map φ is unimodal with a unique nondegenerate critical point, i.e. there exists $\theta^* > \bar{\theta}$ such that $\varphi'(\theta) > 0$ when $\bar{\theta} \leq \theta < \theta^*$, $\varphi'(\theta^*) = 0$ and $\varphi'(\theta) < 0$ when $\theta > \theta^*$. In that case, the conditions $\varphi'(1) < 0$ or $\theta^* < 1$ or $\theta^* < \varphi(\theta^*)$ are equivalent. If anyone of them is satisfied, one has $\theta^* < 1 < \varphi(\theta^*)$ and $\varphi^2(\theta^*) < 1 < \varphi(\theta^*)$.

The proof of the lemma is straightforward. The first part follows trivially from (4.1) and the fact that z_2 is an increasing function. As for the last two points, it suffices to remark that

$$(4.2) \quad \varphi'(\theta) = -z_1'(\theta)/z_2'(\varphi(\theta))$$

which can be rewritten by differentiation of the identity $\theta z_1(\theta) + z_2(\theta) = 0$:

$$(4.3) \quad \varphi'(\theta) = [z_2'(\theta) - (z_2(\theta)/\theta)]/[\theta z_2'(\varphi(\theta))].$$

Thus (4.3) yields $\varphi'(\bar{\theta}) = 1/\bar{\theta} > 1$ since $z_2(\bar{\theta}) = 0$ and $\varphi'(1) < 1$. Finally, the point (iii) and the first part of (iv) are obvious consequences of (4.2) and of Lemma 1.3. The last part of (iv) is true for any unimodal map φ that has $\theta = 1$ as a fixed point.

The map φ seems to be the most natural to consider from an economic viewpoint since it originates from the perfect foresight equilibrium equation for the good market (2.3), which determines equilibrium relative prices and thus real interest rates independently of the money sector. Yet there is nothing intrinsic from a mathematical point of view that is attached to this particular function, and it will be sometimes convenient in the sequel to modify the foregoing dynamical equation $\theta_{t-1} = \varphi(\theta_t)$ by making a *change of variable* of the form $\tilde{\theta} = h(\theta)$ in which the function h is an homeomorphism from $[\bar{\theta}, \infty)$ onto some interval $[a, b)$, i.e. h is continuous and one to one, and h^{-1} is continuous. The equation $\theta_{t-1} = \varphi(\theta_t)$ becomes, then, with the new variable,

$$\tilde{\theta}_{t-1} = h[\varphi(h^{-1}(\tilde{\theta}_t))] \quad \text{or} \quad \tilde{\theta}_{t-1} = \tilde{\varphi}(\tilde{\theta}_t) \quad \text{with} \quad \tilde{\varphi} = h \circ \varphi \circ h^{-1}.$$

The two maps φ and $\tilde{\varphi}$ are then said to be *topological conjugates*. It is clear that they describe the same dynamics: this follows from the relation between the iterates of the two conjugate maps, that is, $\tilde{\varphi}^k = h \circ \varphi^k \circ h^{-1}$. In particular, the two maps have the same cycles up to the change of variable formula $\tilde{\theta} = h(\theta)$, and the stability or instability of a cycle is unaltered by the change of coordinates.²⁴

²⁴ In the particular case where h is a diffeomorphism, i.e. both h and h^{-1} are continuously differentiable and $h'(\theta) \neq 0$ for all θ , it is easy to verify that

$$D\tilde{\varphi}^k(h(\theta)) = h'(\varphi^k(\theta))D\varphi^k(\theta)/h'(\theta).$$

Thus the sign of derivatives is unchanged. Moreover, if θ is a fixed point of φ^k , $h(\theta)$ is a fixed point of $\tilde{\varphi}^k$, and $D\tilde{\varphi}^k(h(\theta)) = D\varphi^k(\theta)$. Finally, it is readily seen by inspection that $\tilde{\varphi}^k(h(\theta)) > h(\theta)$ if and only if $\varphi^k(\theta) > \theta$ (respectively $\varphi^k(\theta) < \theta$) whenever h preserves orientation (respectively reverses orientation), that is when $h'(\theta) > 0$ (respectively $h'(\theta) < 0$) for all θ .

We have already seen an example of topological conjugacy when considering the b.p.f. maps φ and Φ . We shall often use in the sequel a particular change of variable, $h(\theta) = z_2(\theta)$, and shall denote the result $\chi = z_2 \circ \varphi \circ z_2^{-1}$. The function χ has a particularly simple interpretation, for it describes the (backward) perfect foresight dynamics on *equilibrium real balances*. Indeed, we know from (2.4) that these equilibrium real balances μ_t are associated with perfect foresight equilibrium real interest rates θ_t by $\mu_{t+1} = z_2(\theta_t)$. Therefore the equation $\theta_{t-1} = \varphi(\theta_t)$ becomes through this transformation

$$(4.4) \quad \mu_t = z_2(\theta_{t-1}) = \mu_{t+1} / z_2^{-1}(\mu_{t+1}) = \chi(\mu_{t+1}).$$

Of course, since the two maps φ and χ are deduced from each other through the change of variable $\mu = z_2(\theta)$, they must display the same qualitative properties. In particular, if $\alpha_2 = \sup R_2(a_2) \leq 1$, χ is increasing everywhere, while χ has a unique critical point $\mu^* = z_2(\theta^*)$ when $\alpha_2 > 1$ under Assumption (1.e). The transposition of Lemma 4.1 into the corresponding properties of χ is straightforward and is left to the reader. One may remark that the curve representing the map χ is obtained simply from the trader's offer curve by making a rotation of angle $\pi/2$ in the plane (a_1, a_2) around the endowment point (l_1^*, l_2^*) —see Figure 2. *The map χ represents therefore the traders' offer curve.*

It is useful to note that the map φ (or χ) may be defined implicitly by an equation involving marginal utilities. If we look back at the identity (1.7) and apply it for $\theta = \theta_t$, we see that the backward difference equation $z_1(\theta_t) + z_2(\theta_{t-1}) = 0$ is equivalent to

$$(4.5) \quad z_2(\theta_{t-1}) V'_1(l_1^* - z_2(\theta_{t-1})) = z_2(\theta_t) V'_2(l_2^* + z_2(\theta_t))$$

which gives the function in implicit form. By making the change of variable $\mu_{t+1} = z_2(\theta_t)$, (4.5) becomes

$$(4.6) \quad \mu_t V'_1(l_1^* - \mu_t) = \mu_{t+1} V'_2(l_2^* + \mu_{t+1})$$

which defines implicitly the map χ . As a matter of fact, if we define the functions v_1 and v_2 by

$$(4.7) \quad \begin{aligned} v_1(\mu) &= \mu V'_1(l_1^* - \mu) \quad \text{for } \mu \text{ in } [0, l_1^*) \text{ and} \\ v_2(\mu) &= \mu V'_2(l_2^* + \mu) \quad \text{for all } \mu \geq 0, \end{aligned}$$

we see that v_1 is a differentially increasing function that maps the interval $[0, l_1^*)$ onto $[0, +\infty)$, while v_2 maps $[0, +\infty)$ into itself. Then one has $\chi(\mu) = (v_1^{-1} \circ v_2)(\mu)$ for all $\mu \geq 0$. This way of defining the function χ has the advantage of involving directly the traders' characteristics, i.e. their utility functions, instead of their demand functions z_1 and z_2 as in (4.4). This feature will be quite useful in the sequel.

The reader might benefit from relating the properties of the map χ to the corresponding properties of the functions v_1 and v_2 . In particular, if $\alpha_2 = \sup R_2(a_2) \leq 1$, v_2 is increasing everywhere, while v_2 has a unique maximum at $\mu^* = z_2(\theta^*)$ under Assumption (1.e) when $\alpha_2 > 1$. It is moreover easy to verify that

$$(4.8) \quad v_1(\mu) > V'_1(l_1^*)\mu \quad \text{and} \quad v_2(\mu) < V'_2(l_2^*)\mu \quad \text{for all } \mu > 0.$$

This statement, which says in effect that the curve representing v_1 (or v_2) must lie above (or below) its tangent at the origin $\mu = 0$, follows directly from (4.7) and the strict concavity of the utility functions V_r .

Existence and Multiplicity of Periodic Equilibria

We look now at the existence and multiplicity of periodic equilibria. We shall assume throughout Assumptions (1.a), (1.c), (1.d), and shall note as before $\alpha_2 = \sup R_2(a_2)$.

Since $\bar{\theta} < 1$, there is a unique stationary monetary equilibrium $\theta = 1$. We give first conditions under which there is not other cycle with a period $k \geq 2$.

LEMMA 4.2: *Assume (1.a), (1.c), (1.d). If $\alpha_2 \leq 1$, or if Assumption (1.e) holds, $\alpha_2 > 1$ and $z'_2(1) \leq 0$, then the map φ has no cycle with a period $k \geq 2$. Moreover, the unique stationary monetary equilibrium is globally φ -stable, i.e. $\lim_{j \rightarrow +\infty} \varphi^j(\theta) = 1$ for every $\theta > \bar{\theta}$.*

The proof of this statement is immediate. Under the assumptions of the lemma, either φ is increasing everywhere, or φ is unimodal with a unique critical point $\theta^* \geq 1$. Consider now an arbitrary θ_0 that differs from $\bar{\theta}$ and from 1. In the first case, i.e. when $\alpha_2 \leq 1$, then if $\theta_0 < 1$ one has $\theta_0 < \varphi(\theta_0) < \varphi(1) = 1$, and one gets the reverse inequalities whenever $\theta_0 > 1$. By applying the same arguments to the iterates $\varphi^j(\theta_0)$, one gets that $\varphi^j(\theta_0)$ converges monotonically to 1 when j tends to infinity. In the second case, we remark that $\varphi(\theta) \leq \varphi(\theta^*)$ for all θ , so that we may assume without loss of generality that the initial point θ_0 belongs to the interval $(\bar{\theta}, \varphi(\theta^*)]$. Note next that $\theta^* \geq 1$ implies $\varphi(\theta^*) \leq \theta^*$. Thus $\varphi'(\theta) > 0$ for all θ in the interior of the interval $(\bar{\theta}, \varphi(\theta^*)]$. One may therefore reproduce the argument given in the first case to conclude that $\varphi^j(\theta_0)$ converges to 1 as j tends to $+\infty$. This global stability result implies of course the nonexistence of a cycle displaying a period $k \geq 2$.

The foregoing result shows in particular that a necessary condition for the occurrence of a cycle of period $k \geq 2$ (of a “nondegenerate” cycle) is that old traders are sufficiently risk averse ($\alpha_2 > 1$) and that (under Assumption (1.e)) $z'_1(1) > 0$, or equivalently that $\varphi'(1) < 0$. This condition means that the unique critical point of φ satisfies $\theta^* < 1$ (equivalently $\chi(\mu^*) > \mu^*$).

It is straightforward to see that *under Assumptions (1.a), (1.c), (1.d) a sufficient condition for the existence of a cycle of period 2 is that the stationary equilibrium is φ -unstable, i.e. $\varphi'(1) < -1$, or equivalently $z'_1(1) > z'_2(1)$.*²⁵ A cycle of period 2 is indeed described by (θ_1^*, θ_2^*) in which each θ_i^* is a fixed point of the iterate φ^2 , and $\theta_2^* = \varphi(\theta_1^*) \neq \theta_1^*$. It is then clear that in order to find a cycle of period 2, it is necessary and sufficient to find a fixed point $\theta > \bar{\theta}$ of φ^2 that differs from 1—the corresponding orbit being then $(\theta, \varphi(\theta))$. Now the function φ^2 maps the

²⁵ In terms of the functions v_1 and v_2 introduced in (4.7), this condition means $v'_1(z_2(1)) + v'_2(z_2(1)) < 1$. The reader will note also by using (4.3) that the condition $\varphi'(1) < -1$ means that the elasticity of the function z_2 at $\theta = 1$ is less than 0.5, or equivalently that the elasticity of the demand for real balances $z_2(\theta)/\theta$ at $\theta = 1$ is less than -0.5 .

interval $[\bar{\theta}, +\infty)$ into $[\bar{\theta}, \bar{\bar{\theta}})$ and is continuously differentiable. One has $\varphi^2(\bar{\theta}) = \bar{\theta}$, the derivative of φ^2 at $\bar{\theta}$ is equal to $(1/\bar{\theta})^2 > 1$, and of course $\varphi^2(1) = 1$. A sufficient condition for the existence of a fixed point $\theta > \bar{\theta}$ of φ^2 that differs from 1 is therefore that the derivative of φ^2 at $\theta = 1$ is greater than unity, that is

$$(4.9) \quad D\varphi^2(1) = [\varphi'(1)]^2 > 1.$$

Indeed, $\varphi^2(\theta) > \theta$ for $\theta > \bar{\theta}$ provided that θ is close enough to $\bar{\theta}$. Under (4.9), $\varphi^2(\theta) < \theta$ if θ is less than but close enough to 1. Thus by continuity, there exists $\theta \neq \bar{\theta}, 1$ such that $\varphi^2(\theta) = \theta$. Since $\varphi'(1) < 1$, the sufficient condition (4.9) means in effect $\varphi'(1) < -1$, as announced.

It is instructive to reformulate the sufficient condition $z'_1(1) > z'_2(1)$ as follows by using the expressions of the derivatives z'_1 and z'_2 given in Lemma 1.2 (or by using directly the equivalent condition $v'_1(z_2(1)) + v'_2(z_2(1)) < 1$):

$$(4.10) \quad R_2(l_2^* + z_2(1)) > 2 \frac{l_2^* + z_2(1)}{z_2(1)} + \frac{l_2^* + z_2(1)}{l_1^* + z_1(1)} R_1(l_1^* + z_1(1)).$$

This relation suggests that a cycle of period 2 will appear if old traders are sufficiently risk averse. As a matter of fact, the reader will verify by himself that in the constant elasticity case—i.e. $V_\tau(a_\tau) = a^{1-\alpha_\tau}/(1-\alpha_\tau)$ and thus $R_\tau(a_\tau) = \alpha_\tau$ —then (4.10) is satisfied if the old trader's risk aversion is sufficiently large, in the case in which $l_2^* < 1$, $l_2^* < l_1^*$, and $l_1^* + l_2^* > 1$ (hint: find bounds for $z_1(1)$ and $z_2(1)$).

We have seen once again in the course of the previous argument that cycles of period 2 were bound to coexist with the unique stationary monetary equilibrium $\theta = 1$. One may thus expect that cycles displaying different periods will coexist. It turns out that one can be much more precise quite generally about this coexistence. Consider the following ordering of the positive integers:

$$(4.11) \quad \left| \begin{array}{l} 3 > 5 > 7 > \dots \\ > 2 \cdot 3 > 2 \cdot 5 > 2 \cdot 7 > \dots \\ \dots \\ > 2^n \cdot 3 > 2^n \cdot 5 > 2^n \cdot 7 > \dots \\ \dots \\ > \dots > 2^m > \dots > 8 > 4 > 2 > 1. \end{array} \right.$$

That is, consider first the odd integers greater than or equal to 3, then the powers of 2 times these odd integers, and then the powers of 2 backward. We then have the following theorem:²⁶

THEOREM 4.3: *Assume (1.a), (1.c), (1.d). Then, if the map φ has a cycle of period k , it has also a cycle of period k' for every positive integer k' such that $k > k'$.*

²⁶ The proof of this result, which is a direct application of Sarkovskii's beautiful theorem, is given in the Appendix.

Some implications of this very strong result are worth being noted. It shows in particular that cycles with a period 2^m , $m = 0, 1, 2, \dots$ must “appear first” before we have any hope of getting a cycle with an odd period. Moreover, if there is a cycle of period 3, there is also a periodic equilibrium with period n , for every positive integer n .

In the light of the preceding result, it is interesting to investigate the circumstances implying the existence of a cycle of period 3. Such a cycle is represented by its orbit $(\theta_1^*, \theta_2^*, \theta_3^*)$ in which each θ_i^* is a fixed point of φ^3 , $\theta_2^* = \varphi(\theta_1^*) \neq \theta_1^*$ and $\theta_3^* = \varphi^2(\theta_1^*) \neq \theta_1^*$. It is clear here again that a cycle of period 3 is characterized by a fixed point $\theta > \bar{\theta}$ of φ^3 that differs from 1—the corresponding orbit is then described by $(\theta, \varphi(\theta), \varphi^2(\theta))$. Now the function φ^3 maps $[\bar{\theta}, +\infty)$ into $[\bar{\theta}, \bar{\theta})$ and is continuously differentiable. One has $\varphi^3(\bar{\theta}) = \bar{\theta}$, $D\varphi^3(\bar{\theta})$ is equal to $(1/\bar{\theta})^3 > 1$ and $\varphi^3(1) = 1$. One cannot apply here however the argument that we employed for the case of a cycle of period 2, since

$$D\varphi^3(1) = [\varphi'(1)]^3 < 1.$$

But we can state that *under Assumptions (1.a), (1.c), (1.d), a sufficient condition for the occurrence of a cycle of period 3 is that there exists θ such that $\bar{\theta} < \theta < 1$ and $\varphi^3(\theta) \leq \theta$* . Indeed $D\varphi^3(\bar{\theta}) > 1$ implies $\varphi^3(\theta) > \theta$ when θ differs from but is close enough to $\bar{\theta}$, and the result follows by continuity.²⁷ We may then apply the foregoing condition, under the simplifying assumption that φ is unimodal, to its critical point θ^* . Thus, *under the additional Assumptions (1.e) and $\alpha_2 = \sup R_2(a_2) > 1$, a cycle of period 3 will exist provided that $\theta^* < 1$ and $\varphi^3(\theta^*) \leq \theta^*$* (we know from Lemma 4.2 that $\theta^* < 1$ is then a necessary condition for the occurrence of any nondegenerate cycle). It is worth noting that under such a circumstance, one has necessarily

$$\varphi^2(\theta^*) < \varphi^3(\theta^*) \leq \theta^* < 1 < \varphi(\theta^*).$$

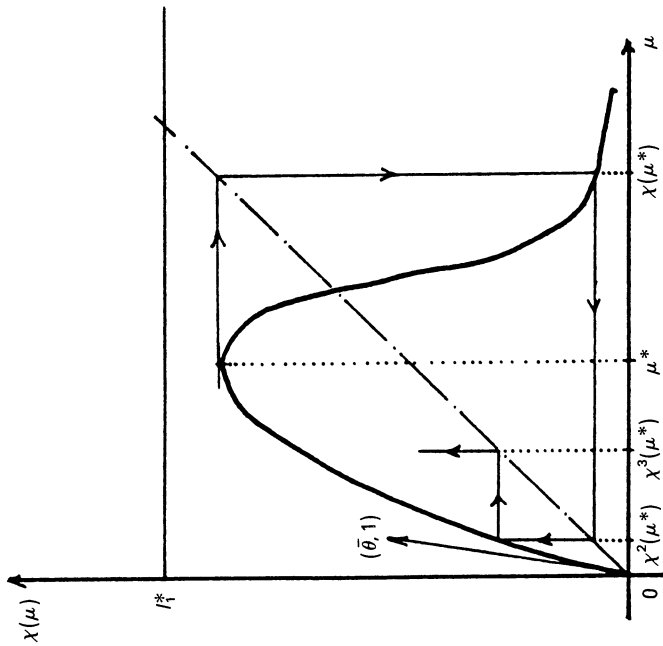
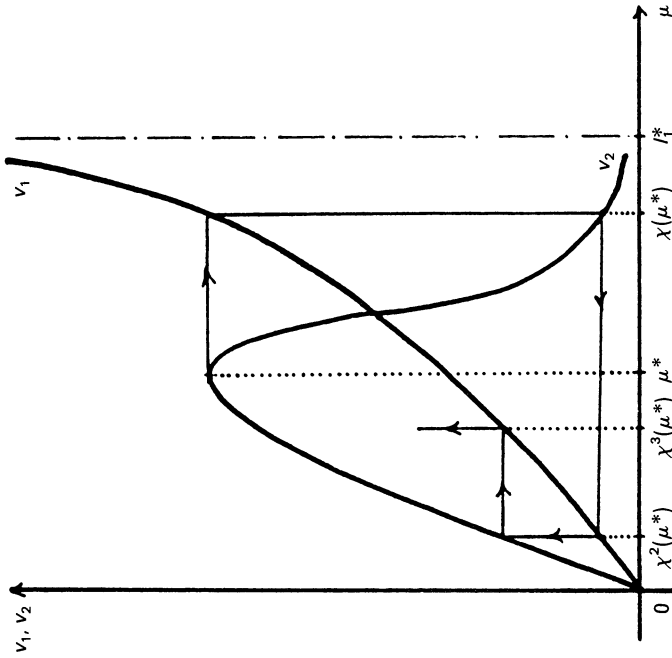
Indeed we know from Lemma 4.1(iv) that $\theta^* < 1$ implies $\varphi^2(\theta^*) < 1 < \varphi(\theta^*)$, in which case $\varphi^2(\theta^*) < \varphi^3(\theta^*)$, holds also from (i) of the same lemma.

What we just said for the map φ is of course valid for any of its topological conjugates, and thus for $\chi = v_1^{-1} \circ v_2$: under the above Assumptions (1.a), (1.c), (1.d), (1.e) and $\alpha_2 > 1$, a cycle of period 3 occurs provided that $\chi(\mu^*) > \mu^*$ and $\chi^3(\mu^*) \leq \mu^*$. In that case we have also of course

$$\chi^2(\mu^*) < \chi^3(\mu^*) \leq \mu^* < z_2(1) < \chi(\mu^*).$$

Figure 3.a describes a case in which the phenomenon obtains. The central feature of the example is, as one might have expected, that the offer curve displays an important “hump”. Figure 3.b represents the implications of the example for

²⁷ The condition is also *necessary*. For otherwise the orbit $(\theta_1^*, \theta_2^*, \theta_3^*)$ of the cycle would satisfy $\theta_i^* > 1$ and thus $\theta_i^* > \varphi(\theta_i^*)$. That would imply $\theta_1^* > \theta_2^* > \theta_3^* > \varphi(\theta_3^*)$, a contradiction. The same reasoning shows that under (1.a), (1.c), (1.d), a necessary and sufficient condition for the existence of a cycle of period 3 is that $\varphi^3(\theta) \geq \theta$ for some $\theta > 1$. The reader will note incidentally that the map φ must have generically an even number of cycles of period 3. This comes from the fact that, generically, the number of solutions of $\varphi^3(\theta) = \theta$ in $(\bar{\theta}, +\infty)$ must be odd.



the respective shapes of the curves corresponding to the functions v_1 and v_2 (the Figure shows incidentally also how the dynamics associated with the equation $v_1(\chi(\mu)) = v_2(\mu)$ may be described with the help of the curves v_1 and v_2). Again, what is important is that v_2 displays a large hump, so that $v_2(\mu^*)$ is “large” while $v_2(\chi(\mu^*))$ is “small.”

We wish now to translate these heuristic statements into a formal condition that involves only the traders’ utility functions, or equivalently the functions v_1 and v_2 . Let us assume accordingly (1.a), (1.c), (1.d). Then under Assumption (1.e) and $\alpha_2 > 1$, the function v_2 has a unique maximum which occurs at the critical point μ^* of $\chi = v_1^{-1} \circ v_2$. The condition $\mu^* < \chi(\mu^*)$ means in effect that $v_2(\mu^*) > v_1(\mu^*)$. Since v_1 is increasing, the inequality $\mu^* < \chi(\mu^*)$ will thus obtain if and only if there exists $\bar{\mu} \geq \mu^*$ such that $v_2(\mu^*) > v_1(\bar{\mu})$. In this case we have $\mu^* \leq \bar{\mu} < \chi(\mu^*)$ and since $\chi'(\mu) < 0$ for all $\mu > \mu^*$ we get that $\chi^2(\mu^*) < \chi(\bar{\mu})$. On the other hand, since the curve χ lies below its tangent at $\mu = 0$, i.e. $\chi(\mu) < \mu/\bar{\theta}$ for all $\mu > 0$,²⁸ we have

$$\chi^3(\mu^*) < \chi^2(\mu^*)/\bar{\theta} < \chi(\bar{\mu})/\bar{\theta}.$$

The condition $\chi^3(\mu^*) \leq \mu^*$ will obtain accordingly (in fact with a strict inequality sign) if

$$\chi(\bar{\mu}) = (v_1^{-1} \circ v_2)(\bar{\mu}) \leq \bar{\theta}\mu^*$$

or equivalently if

$$(4.12) \quad v_2(\bar{\mu}) \leq v_1(\bar{\theta}\mu^*).$$

Since the curve representing the function v_1 must lie above its tangent at the origin (see (4.8)), we have $v_1(\bar{\theta}\mu^*) > V_1'(l_1^*)\bar{\theta}\mu^*$, and thus (4.12) will automatically be satisfied if

$$(4.13) \quad v_2(\bar{\mu}) \leq V_1'(l_1^*)\bar{\theta}\mu^*.$$

To sum up, we have obtained Proposition 4.4.²⁹

PROPOSITION 4.4: Assume (1.a), (1.c), (1.d), (1.e), and $\alpha_2 = \sup R_2(a_2) > 1$. Let v_1 and v_2 be defined by (4.7) and let μ^* be the unique maximum of v_2 . Assume that there exists $\bar{\mu} \geq \mu^*$ such that $v_2(\mu^*) > v_1(\bar{\mu})$, that satisfies $v_2(\bar{\mu}) \leq v_1(\bar{\theta}\mu^*)$ or the stronger condition $v_2(\bar{\mu}) \leq V_1'(l_1^*)\bar{\theta}\mu^*$. Then one has³⁰

$$\chi^2(\mu^*) < \chi^3(\mu^*) < \mu^* < \chi(\mu^*)$$

and there exists a cycle of period 3.

²⁸ This follows directly from (4.4) and the fact that $z_2^{-1}(\mu) > \bar{\theta}$ whenever $\mu > 0$. One can also argue directly that the curve χ is isometric to the trader’s offer curve.

²⁹ The conditions that we get for the existence of a cycle of period 3 are closely related to the conditions obtained by Benhabib and Day [7] in their study of the occurrence of “chaotic” behavior in an overlapping generations model.

³⁰ The same statement could have been made of course by using φ and its critical point θ^* .

Apart from providing a criterion to verify whether a cycle of period 3 exists, the foregoing Proposition gives a way to “generate” utility functions that entail the appearance of a cycle of period 3. Choose μ^* and $\bar{\mu}$ such that $\mu^* < \bar{\mu} < l_1^*$. We then keep fixed all the characteristics of the model, except the old trader’s utility function V_2 (or v_2) which we shall vary subject to the restriction that (1.e) holds, and that v_2 reaches its maximum at μ^* . Choose a value of $V_2'(l_2^* + \mu^*)$ that is sufficiently high to satisfy

$$\mu^* V_2'(l_2^* + \mu^*) > \bar{\mu} V_1'(l_1^* - \bar{\mu})$$

and pick an arbitrary value of $V_2'(l_2^*)$ ($> V_2'(l_2^* + \mu^*)$). Then it suffices to impose a value of $V_2'(l_2^* + \bar{\mu})$ that is sufficiently low to satisfy (4.13). The only restriction is in fact to fulfill Assumption (1.e).

The next result confirms the intuition that we had when studying the occurrence of cycles of period 2, namely that the more a trader’s relative degree of risk aversion increases as he gets older, the more likely is the emergence of cycles.

COROLLARY: *Consider the constant elasticity case, that is $V_\tau(a_\tau) = a_\tau^{(1-\alpha_\tau)}/(1-\alpha_\tau)$ in which case $R_\tau(a_\tau) = \alpha_\tau > 0$. Assume $0 < l_2^* < 1$ and keep fixed all the characteristics of the model except the old trader’s relative risk aversion α_2 that is free to vary. Then if $l_1^* + l_2^* > (1/l_2^*)$, the assumptions of Proposition 4.4 are verified, and a cycle of period 3 thus exists, when α_2 is large enough.*

The proof of this assertion uses only elementary algebra. In the particular case under consideration, we have

$$v_1(\mu) = \mu(l_1^* - \mu)^{-\alpha_1} \quad \text{and} \quad v_2(\mu, \alpha_2) = \mu(l_2^* + \mu)^{-\alpha_2}.$$

We have then $\bar{\theta}(\alpha_2) = (l_2^*)^{\alpha_2}/(l_1^*)^{\alpha_1}$ which is a decreasing function of α_2 and goes to 0 as α_2 tends to infinity, since $l_2^* < 1$. Remark next that when μ is fixed and α_2 diverges to infinity, $v_2(\mu, \alpha_2)$ tends to infinity when $0 < \mu < 1 - l_2^*$, to 0 when $\mu > 1 - l_2^*$, and that $v_2(1 - l_2^*, \alpha_2) = 1 - l_2^*$ for all α_2 . On the other hand, when $\alpha_2 > 1$, the maximum of v_2 with respect to μ occurs at $\mu^*(\alpha_2) = l_2^*/(\alpha_2 - 1)$, which is a decreasing function of α_2 and goes to 0 as α_2 tends to infinity. It is not difficult then to verify that the maximum of v_2 , that is $v_2(\mu^*(\alpha_2), \alpha_2)$, diverges to infinity when α_2 goes to $+\infty$.

If we have $l_1^* + l_2^* > (1/l_2^*)$, then $l_1^* + l_2^* > 1$ since $l_2^* < 1$. Choose now an arbitrary $\bar{\mu}$ such that $1 - l_2^* < \bar{\mu} < l_1^*$. We have clearly $\mu^*(\alpha_2) < \bar{\mu}$ and $v_2(\mu^*(\alpha_2), \alpha_2) > v_1(\bar{\mu})$ for α_2 large enough. On the other hand, (4.13) will be satisfied if one can choose $\bar{\mu}$ so that

$$v_2(\bar{\mu}, \alpha_2)/[V_1'(l_1^*)\bar{\theta}(\alpha_2)\mu^*(\alpha_2)] \leq 1$$

for α_2 large enough. But it is not difficult to verify that the left hand member of this inequality goes actually to 0 when α_2 tends to $+\infty$ if $\bar{\mu} + l_2^* > (1/l_2^*)$. Thus if we pick up $\bar{\mu}$ such that $1 - l_2^* < \bar{\mu} < l_1^*$ and $\bar{\mu} + l_2^* > (1/l_2^*)$, which is always possible when $l_1^* + l_2^* > (1/l_2^*)$, then the assumptions of Proposition 4.4 are fulfilled when α_2 is large, as claimed.

Uniqueness of φ -stable Cycles

Theorem 4.3 above shows that cycles of very different periods will typically coexist. We now present a condition which ensures that there exists *at most* one φ -stable cycle. The condition is essentially that the map φ , or one of its topological conjugates, is unimodal and has a “Schwarzian derivative” (to be defined shortly) that is negative. This condition is there to guarantee that the map φ has good “expansive” properties. It will obtain in particular, in the constant elasticity case, when traders have a degree of risk aversion less than or equal to 1 in their youth and greater than or equal to 2 when they are older. Then all cycles of φ with the exception of at most one, are unstable in the b.p.f. dynamics and are thus stable in the local forward perfect foresight dynamics as defined in Section 3.

On the other hand, if the traders’ expectation function is consistent with the periodicity of the unique φ -stable cycle, if there is any, then this cycle is also stable in the dynamics with learning under Assumption (3.h) (Proposition 3.2). The latter dynamics may however have other stable cycles.

We shall need a refinement of the definition of stability for the purpose of the present study. A cycle $(\theta_k^*, \dots, \theta_1^*)$ of φ will be said to be φ -weakly stable if $D\varphi^k(\theta_i^*) \leq 1$. If φ is unimodal, the cycle will be φ -superstable if the critical point of φ belongs to the periodic orbit, that is if $D\varphi^k(\theta_i^*) = 0$. The notion of φ -stability is defined of course as before.

The object of the analysis that follows is to study the conditions under which the map φ has at most one weakly stable periodic orbit, under the assumption that φ is unimodal. Of course, in view of Lemma 4.2, in order to make the problem nontrivial, we need to assume, in addition to (1.a), (1.c), (1.d), (1.e), that $\alpha_2 = \sup R_2(a_2) > 1$ and that $\varphi'(1) < 0$, or equivalently that the critical point of φ satisfies $\theta^* < 1$.

The result that we shall present uses the notion of a “Schwarzian derivative,” which we define now. If f is a thrice continuously differentiable function that maps the interval $[a, b]$ of the real line into itself, then the Schwarzian derivative³¹ of f , noted Sf , is defined for every x in $[a, b]$ such that $f' \neq 0$ by

$$(4.14) \quad Sf = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2.$$

Direct inspection shows that $Sf = -2|f'|^{1/2} D^2[|f'|^{-1/2}]$. So the condition that “ f has a negative Schwarzian derivative” ($Sf < 0$ at every x such that $f'(x) \neq 0$) means that $|f'|^{-1/2}$ is convex on every interval of monotony of f . It will be satisfied in particular if $|f'|$ (or $\log |f'|$) is concave on such intervals. But these sufficient conditions are by no means necessary. Finally the reader will note that the concavity of f is neither necessary nor sufficient to guarantee $Sf < 0$.

In order to use this notion, we must ensure that φ and thus the excess demand functions z_1 and z_2 are thrice continuously differentiable. This is achieved if we reinforce (1.c) by adding to it the following condition:

³¹ The role of the Schwarzian derivative in the study of one-dimensional dynamical systems has been discovered by Singer [52].

ASSUMPTION (4.c'): *The indirect utility functions V_τ have continuous third and fourth derivatives on $(0, +\infty)$.*

It is then clear that all topological conjugates of φ obtained through a change of variable $\bar{\theta} = h(\theta)$ will be also thrice continuously differentiable provided that h has continuous third derivatives. This will be in particular the case of the map χ .

Then we have the following important result.

THEOREM 4.5: *Assume (1.a), (1.c), (1.d), (1.e), (4.c'), $\alpha_2 = \sup R_2(a_2) > 1$, and that the critical point of φ satisfies $\theta^* < 1$. Assume moreover that φ (or one of its topological conjugates $\tilde{\varphi}$) has a negative Schwarzian derivative on the interval $[\bar{\theta}, \varphi(\theta^*)]$ (or on the corresponding interval $[h(\bar{\theta}), h(\varphi(\theta^*))]$). Then (i) The map φ has at most one weakly stable periodic orbit; (ii) if φ has a weakly stable periodic orbit, then the critical point θ^* is attracted to it, that is, this periodic orbit coincides with the set of accumulation points of the sequence $(\varphi^j(\theta^*))$.³²*

Besides the uniqueness of φ -weakly stable periodic orbits, the foregoing result gives an “experimental” way of establishing whether or not a particular map φ has a weakly stable cycle provided that it has (itself or one of its topological conjugates) a negative Schwarzian derivative. It suffices to iterate the critical point by using a computer, to check whether or not the iterates converge and to verify that the limit cycle, if any, is indeed weakly stable. Of course, since the iterations must be stopped in practice after some time, this procedure discloses cycles that have a small period, but is not able to distinguish between the presence of a weakly stable cycle that has a long period and the absence of any weakly stable periodic orbit. Finally it should be emphasized that *there are maps that do not have any weakly stable periodic orbit*. Such maps are called *aperiodic*. The foregoing result tells us that a map φ that has a negative Schwarzian derivative will be aperiodic whenever the iterates of θ^* do not converge or if they converge to an *unstable* cycle.

There is an obvious difficulty in applying Theorem 4.5 as it is stated, for the negativity of the Schwarzian derivative, like any statement involving convexity is not invariant when making a (nonlinear) change of variable. We may get accordingly from a particular economic model a b.p.f. map φ that does not satisfy $S\varphi < 0$ whereas one of its topological conjugates does. In that case, Theorem 4.5 applies but the model builder may not be able to diagnose it by looking at the map φ . So it is important to have a criterion using the basic characteristics of the model (i.e. utility functions and endowments) that allows recognizing directly that Theorem 4.5 applies. A partial answer to this question is provided by the following lemma:

³² The proof of this statement, which is a direct application of the results reported in Collet and Eckmann [11], is given in the Appendix.

LEMMA 4.6: Assume (1.a), (1.c), (1.d), (1.e), (4.c'), $\alpha_2 = \sup R_2(a_2) > 1$, and that the critical point of φ satisfies $\theta^* < 1$. Consider the functions v_1 and v_2 defined in (4.7) and assume that $Sv_1(\mu) \geq 0$ for all μ in $[0, l_1^*)$ and $Sv_2(\mu) < 0$ on the interval $[0, \chi(\mu^*)]$. Then $S\chi < 0$ on $[0, \chi(\mu^*)]$ and Theorem 4.5 applies.

In particular, the foregoing condition on v_1 and v_2 obtains in the constant elasticity case— $V_\tau(a_\tau) = a_\tau^{(1-\alpha_\tau)}/(1-\alpha_\tau)$ and thus $R_\tau(a_\tau) = \alpha_\tau > 0$ —if $0 < \alpha_1 \leq 1$ and $\alpha_2 \geq 2$.

The proof of this statement uses the fact which may be verified by direct computation that the Schwarzian derivative of the composition of two (thrice continuously differentiable) maps f and g is given by

$$(4.15) \quad S(f \circ g)(x) = Sf(g(x))[g'(x)]^2 + Sg(x).$$

Applying (4.15) to $v_1 \circ \chi = v_2$, we get for all $\mu \geq 0$,

$$Sv_2(\mu) = Sv_1(\chi(\mu))[\chi'(\mu)]^2 + S\chi(\mu).$$

The first part of the lemma is then immediate.

In the case of a constant relative risk aversion, one has $v_1(\mu) = \mu(l_1^* - \mu)^{-\alpha_1}$ and $v_2(\mu) = \mu(l_2^* + \mu)^{-\alpha_2}$. Since $v_1'(\mu) > 0$ for all μ in $[0, l_1^*)$, Sv_1 is defined everywhere on that interval. Direct computation shows that Sv_1 is equal to (up to an everywhere positive factor)

$$(1 - \alpha_1)[(1 - \alpha_1)(2 - \alpha_1)\mu^2 - 4l_1^*(2 - \alpha_1)\mu + 6(l_1^*)^2].$$

The expression between the brackets is easily seen to be a monotone function of μ on the interval $[0, l_1^*)$, which is positive on this interval. Thus $Sv_1 > 0$ on $[0, l_1^*)$ if and only if $\alpha_1 < 1$. As for v_2 , Sv_2 is defined for every $\mu \neq \mu^*$ on $[0, +\infty)$ and the same sort of computation shows that it has the same sign as

$$-(\alpha_2 - 1)[(\alpha_2 - 1)(\alpha_2 - 2)\mu^2 - 4l_2^*(\alpha_2 - 2)\mu + 6(l_2^*)^2].$$

It is not difficult to verify that the expression between the brackets is positive for every $\mu \geq 0$ when $\alpha_2 \geq 2$, which establishes the Lemma.

REMARK: Under the assumptions of Theorem 4.5, it is possible to show that if the dynamical system φ has a weakly stable periodic orbit, then the set of points that are not attracted to it is “exceptional,” i.e. has Lebesgue measure zero (this follows from Collet and Eckmann [16, Proposition II.5.7]). We do not insist on this otherwise nice result for the backward perfect foresight dynamics has no clear meaning in the present context, beyond the fact that φ -stability implies stability in the dynamics with learning under Assumption (3.h). We mention the result nonetheless since it shows that some claims that “period three implies chaos” are generally unwarranted. In particular, Benhabib and Day [6, 7], Day [18, 19] use a result of Li and Yorke [41], or a variant of it, to exhibit, under the assumption that there is a cycle of period 3, a “chaotic set,” that is a set such that any perfect foresight trajectory starting from it becomes eventually erratic.

This discussion shows the limits of such a statement. For if there is a weakly stable periodic orbit, the “chaotic set”—of which the existence is rightly asserted—may well be of Lebesgue measure zero, and erratic behavior may thus be essentially unobservable—see also the remarks in Collet and Eckmann [16, p. 20].

Bifurcation of φ -stable Periodic Equilibria

We consider now the following experiment. Suppose that we take a “one-dimensional” family of economies. That is, we index the characteristics of the economy by a real number and then move this parameter over the real line. Sarkovskii’s theorem suggests that the emergence of cycles as the parameter moves on should display some regularities. We show below that it is indeed the case for φ -stable cycles, and that these regularities are in particular very strong for those φ -stable cycles that should “appear first” according to the Sarkovskii’s ordering (4.11), namely cycles that have a period equal to a power of 2. Finally, we shall report on a computer experiment that was done in the case of a constant relative risk aversion, in which the old traders’ risk aversion is varied, in fact increased. The result of this experiment is a “period doubling” bifurcation scheme very much like the diagrams that are traditional in the analysis of nonlinear one-dimensional dynamical systems.

In order to go on we must make precise what we mean by a (*one-parameter*) *family of economies*. Let us index the characteristics of the economy (the endowments l_τ^* and the indirect utility functions V_τ) by some parameter λ , which will be taken as a real number that belongs to say, the interval $[0, 1]$. The result of this indexation is denoted $l_{\tau\lambda}^*$ and $V_{\tau\lambda}(a_\tau)$, $\tau = 1, 2$. We assume of course that for each λ the characteristics of the economy satisfy Assumptions (1.a), (1.c), (1.d), and in order to make the problem nontrivial, also Assumption (1.e) as well as $\alpha_{2\lambda} = \sup_{a_2} R_{2\lambda}(a_2) > 1$. The corresponding backward perfect foresight map φ_λ has then a unique critical point for each λ , say θ_λ^* , and we shall postulate that $\theta_\lambda^* < 1$ for all λ (again, this condition is necessary in view of Lemma 4.2 to get nondegenerate cycles). We shall say finally that the family is *continuous* if in addition to these assumptions, the endowments $l_{\tau\lambda}^*$ depend continuously on λ and if $V'_{\tau\lambda}(a_\tau)$ as well as $V''_{\tau\lambda}(a_\tau)$ are *jointly* continuous in (a_τ, λ) for $\tau = 1, 2$.

In order to state the results most clearly, it will be convenient to consider a family of economies that is “rich” enough. We shall say accordingly that the family is *full* if it satisfies the two following conditions:

1. When $\lambda = 0$, the second iterate of the critical point θ_0^* is such that $\theta_0^* < \varphi_0^2(\theta_0^*)$. In other words, since from Lemma 4.1(iv) we have already $\varphi_0^2(\theta_0^*) < 1 < \varphi_0(\theta_0^*)$, and since φ_0 is unimodal, all points of the trajectory of the critical point θ_0^* are greater than θ_0^* and oscillate around the stationary equilibrium $\theta = 1$. It can be shown that under this condition, then if the map φ_0 has a nondegenerate cycle, it must have a period at most equal to 2.³³

³³ For details, see Collet and Eckmann [16], especially Lemma II.2.12.

2. When $\lambda = 1$, the map φ_1 satisfies, for instance, the assumptions of Proposition 4.4, so that the trajectory of the critical point θ_1^* fulfills

$$\varphi_1^2(\theta_1^*) < \varphi_1^3(\theta_1^*) < \theta_1^* < \varphi_1(\theta_1^*)$$

and there exists a cycle of period 3.

The following result gives some insight about the fashion in which stable cycles emerge when the index λ of a continuous full family of economies moves from 0 to 1 (in the remainder of this section, “stability” will always refer to the b.p.f. dynamics associated with the maps φ_λ , unless otherwise specified). First, *superstable* cycles of *all* periods $k \geq 2$ do occur for some values of the parameter λ . Of course, given a value of λ for which a superstable cycle obtains, a stable cycle of the same period should still exist by continuity when the characteristics of the economy are changed only a little. Second, stable periods that are equal to a power of 2 appear first and in a *consecutive* manner, in conformity to the ordering (4.11) of the integers. So if the family is “nice”, one should observe a neat pattern of “period doubling” bifurcations, a stable cycle of period 2^j becoming unstable and giving rise to a cycle of period 2^{j+1} , and so on. Such a monotonic pattern is not general however (think of a nonmonotonic reparametrization of the family) and the sequence of stable periods 2^j may be “visited” consecutively but in a *nonmonotonic* fashion as the index λ increases.

Nevertheless, the general scheme is that the sequence of values λ_j^* for which a superstable cycle of period 2^j appears *first*, increases and converges to some value $\lambda_\infty^* < 1$. Thus period doubling bifurcations intervene more and more rapidly as λ moves closer to λ_∞^* . The nice point is that if for each λ the map φ_λ (or one of its topological conjugates) has a negative Schwarzian derivative as in Theorem 4.5, then there exists a unique weakly stable cycle—the period of which is a power of 2—for every $\lambda < \lambda_\infty^*$. In that case weakly stable cycles with a period that differs from a power of 2 must obtain only in the interval $(\lambda_\infty^*, 1]$. In fact a lot of phenomena may occur in that region. In particular there is an uncountable set of values of λ in $(\lambda_\infty^*, 1]$ for which the map φ_λ has no weakly stable periodic orbit.

Formally, we have Theorem 4.7:³⁴

THEOREM 4.7: *Consider a full continuous family of economies.*

(i) *Given an arbitrary integer $k \geq 2$, the set of parameters λ for which the map φ_λ has a superstable cycle of period k is closed and nonempty. Given such a λ , there is an open interval around λ such that $\varphi_{\lambda'}$ has a stable cycle of period k for all λ' in the interval.*

(ii) *Let λ_j^* be the first value of the parameter λ for which a superstable cycle of period 2^j obtains for $j \geq 1$.³⁵ Then the sequence λ_j^* increases with j and converges*

³⁴ The proof of this theorem which is a direct application of the results reported in Collet and Eckmann [16] is given in the Appendix. There are a lot of other nice results on the regularities of the bifurcations of a unimodal map φ which are beyond the scope of this paper. For more information, see Collet and Eckmann.

³⁵ That is $\lambda_j^* = \min \{\lambda \mid \varphi_\lambda \text{ has a superstable cycle of period } 2^j\}$.

to some value $\lambda_\infty^* < 1$ as j tends to $+\infty$. For each λ in $[0, \lambda_\infty^*)$, all cycles of the map φ_λ have a period that is a power of 2 or are fixed points. The critical point θ_λ^* of φ_λ is attracted to one of these.

(iii) If superstable cycles of periods 2^j and $2^{j'}$ with $j' > j + 1$ occur respectively for the values λ and λ' in $[0, \lambda_\infty^*)$, then a superstable cycle of period 2^i with $j' > i > j$ must appear for some value in the open interval determined by λ and λ' .

(iv) Assume that each φ_λ (or one of its topological conjugates) satisfies (4.c') and has a negative Schwarzian derivative. Then for every λ in $[0, \lambda_\infty^*)$, φ_λ has a (unique) weakly stable periodic orbit.

(v) Under the assumptions of (iv), there exists an uncountable set of values of λ in $(\lambda_\infty^*, 1]$ such that φ_λ has no weakly stable cycle.

The foregoing result suggest that one should observe first regular period doubling bifurcations in a one-dimensional full continuous family of economies on some interval, and then that chaotic behavior as well as stable cycles with a period that differ from a power of 2 should obtain later, perhaps in a mixed way. This is in fact the pattern that is always observed in computer simulations of one-dimensional dynamical systems (see May [48], or Collet and Eckmann [16, p. 26]).

The following diagram (Figure 4) shows the result of a similar computer experiment³⁶ implemented in our context in the case of a constant relative degree of risk aversion, i.e. $V_\tau(a_\tau) = a_\tau^{(1-\alpha_\tau)}/(1-\alpha_\tau)$ and $R_\tau(a_\tau) = \alpha_\tau > 0$, in which the parameter indexing the family has been taken to be the old traders' relative risk aversion α_2 . Then we know from Lemma 4.6 that if $\alpha_1 \leq 1$ and $\alpha_2 \geq 2$, Theorem 4.5 applies. In that case, it suffices to iterate the critical point θ^* sufficiently long to discover the unique weakly stable cycle whenever it exists. This procedure was applied in fact to the map χ that describes the backward perfect foresight dynamics of equilibrium real balances. The parameter α_2 was made to vary between 2 and 16 by steps of 0.05.

For each such value of α_2 on the horizontal axis, the corresponding map χ was iterated 300 times, the initial point being the critical point of the map, which is in that case $\mu^* = l_2^*/(\alpha_2 - 1)$. On the vertical axis above α_2 were plotted all the values of the iterated equilibrium real balances μ_t from $t = 200$ to $t = 300$. With this procedure, one may hope the figure shows clearly the weakly stable cycles that do not have too long a period and that are attractive enough. The experiment cannot of course discriminate between stable cycles that have a long period, and erratic behavior. The result of the experiment is shown below in Figure 4, under the specification $\alpha_1 = 0.5$, $l_1^* = 2$, and $l_2^* = 0.5$. It shows indeed the usual period doubling bifurcation pattern.

The general lesson of this discussion is that *when the traders' offer curve becomes more and more humpy, more and more cycles appear that are unstable in the b.p.f.*

³⁶ The computer experiment has been in fact run by Rose-Anne Dana and Dominique Lévy, whose help is gratefully acknowledged. Earlier computer simulations by Alain Morineau were also very valuable.

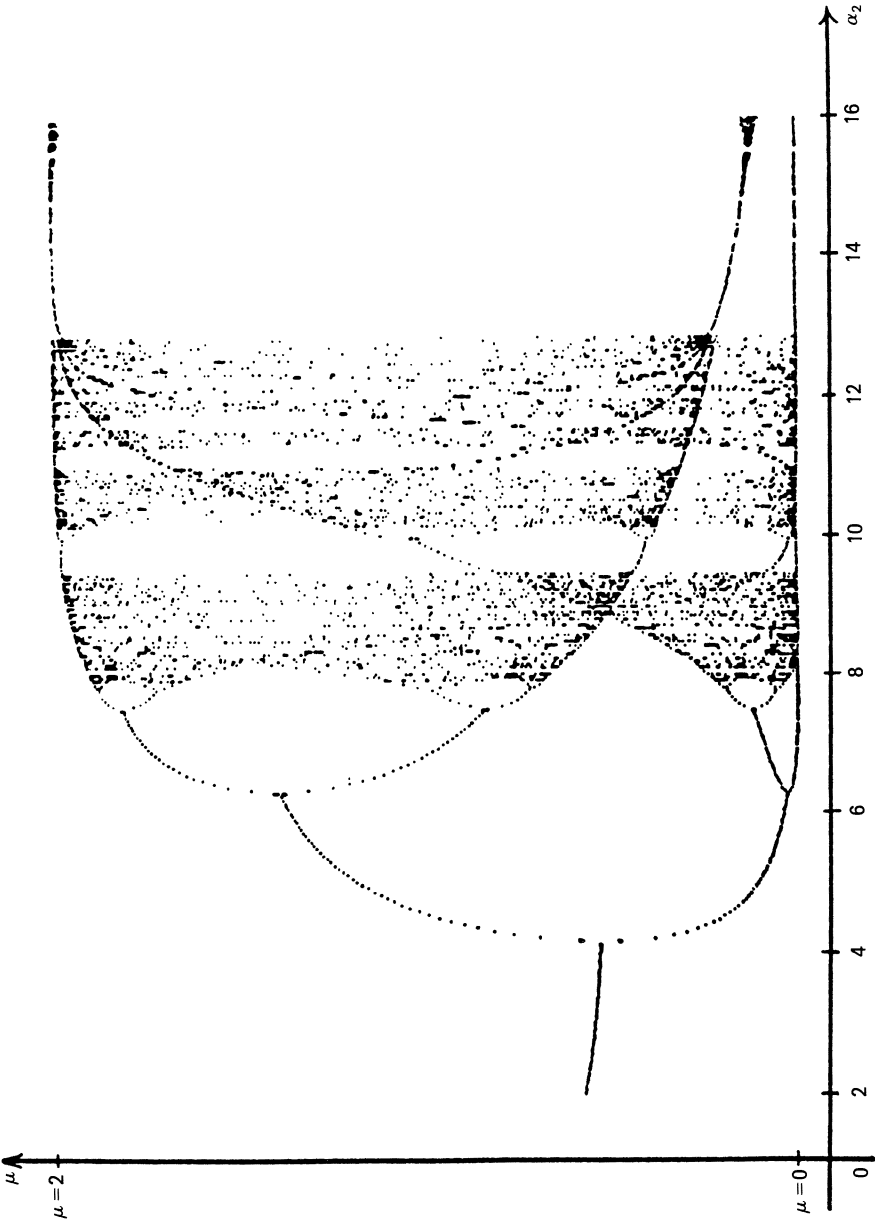


FIGURE 4

dynamics, or equivalently stable in the local forward perfect foresight dynamics, as defined in Section 3.

5. OUTPUT AND ANTICIPATED (OR UNANTICIPATED) REAL INTEREST RATES

The proposition that only unanticipated inflation matters has belonged to the folklore of the “New Classical” macroeconomics over the last decade. We wish now to look more closely at this issue. Since there are movements of output (or equivalently of the rate of participation of the labor force) and of prices along a sequence of competitive equilibria, we may ask whether there is any systematic relationship between the equilibrium levels of output and of the real rate of interest. It will be shown that such a relation exists indeed in the present model, *both in the short run (on the transition path) and in the long run (along periodic equilibria)*. Of course, since no interest is paid on money in the model, this implies a relation between equilibrium output and inflation that goes simply in the opposite direction. It will be seen moreover that high levels of output are associated with low real interest rates (i.e. with high levels of inflation) whenever old traders have a higher marginal propensity to consume leisure than young consumers. Output and inflation are then “procyclical.” Furthermore, it will be shown that such an association holds for anticipated as well as unanticipated movements of the real interest rate (or the rate of inflation).

Assume (1.a), (1.b), (1.c), and consider a sequence of temporary competitive equilibria as described by the conditions (2.1), (2.2). In fact, in view of (1.2), the money market equation yields the following relation between equilibrium real balances at t , i.e. $\mu_t = M/p_t$, and the price p_t^e that was expected for the current date by the young trader at date $t-1$:

$$\mu_t = p_t^e z_2(p_{t-1}/p_t^e)/p_t$$

or equivalently

$$(5.1) \quad \mu_t = \frac{p_t^e}{p_{t-1}} \frac{p_{t-1}}{p_t} z_2\left(\frac{p_{t-1}}{p_t^e}\right).$$

We see that the equilibrium real balance μ_t at t is a function of the real interest rate that was expected in equilibrium at date $t-1$, or more precisely of $\theta_{t-1}^e = p_{t-1}/p_t^e$, and of the forecasting error actually made on this variable, i.e. of $e_t = \theta_{t-1}/\theta_{t-1}^e$, in which $\theta_{t-1} = p_{t-1}/p_t$ (one has equivalently $e_t = p_t^e/p_t$). Indeed, (5.1) can be rewritten with this notation:

$$(5.2) \quad \mu_t = e_t z_2(\theta_{t-1}^e).$$

Since the nominal interest rate paid on money is zero in the model, there is an inverse relationship between the (expected) rate of inflation and the (expected) real rate of interest. In particular, a high level of “unanticipated inflation” means that the real interest rate θ_{t-1} is much lower than was expected, that is, it means a low value of e_t . Of course, perfect foresight corresponds to $e_t = 1$ and thus to $\theta_{t-1}^e = \theta_{t-1}$ (or $p_t^e = p_t$).

The relation (5.2) implies that there is a positive association, along any sequence of temporary equilibria, between real money balances and anticipated or unanticipated real interest rates. Along periodic equilibria with perfect foresight, the relation boils down to $\mu_t = z_2(\theta_{t-1})$, which is of course the same as (2.4).

It is now straightforward to see that (5.2) implies a similar relationship between equilibrium output and anticipated or unanticipated real interest rates. The sequence of real equilibrium money stocks $\mu_t = M/p_t$ satisfies $0 < \mu_t < l_1^*$ for all t . At date t , the young trader's equilibrium consumption and labor supply are then obtained by maximizing $U_1(c_1, l_1^* - l_1)$ under the budget constraint $[c_1 + (l_1^* - l_1)] = l_1^* - \mu_t$. In view of problem (1.4), the outcome is $c_1(l_1^* - \mu_t)$ and $l_1(l_1^* - \mu_t)$. By the same kind of reasoning, the old trader's equilibrium consumption and labor supply are given by $c_2(l_2^* + \mu_t)$ and $l_2(l_2^* + \mu_t)$. Equilibrium output y_t at date t is therefore

$$(5.3) \quad y_t = l_1(l_1^* - \mu_t) + l_2(l_2^* + \mu_t) \\ = c_1(l_1^* - \mu_t) + c_2(l_2^* + \mu_t).$$

Examination of (5.2), (5.3) shows that along an arbitrary sequence of temporary competitive equilibria, equilibrium output is linked to the expected real interest rate θ_{t-1}^e and the forecasting error made on this variable $e_t = \theta_{t-1}/\theta_{t-1}^e$ by $y_t = F(\theta_{t-1}^e, e_t)$, in which

$$(5.4) \quad F(\theta, e) = l_1(l_1^* - ez_2(\theta)) + l_2(l_2^* + ez_2(\theta)),$$

the domain of definition of F being the set of pairs (θ, e) such that $\bar{\theta} \leq \theta \leq z_2^{-1}(l_1^*/e)$ in order to ensure that $0 \leq ez_2(\theta) \leq l_1^*$.

Two important facts are worth noticing. First, the “short run statistical relationship” described by the map F depends only on the real characteristics of the economy, and not on the level of the money stock M . More importantly, *it does not depend on the particular trajectory that is considered nor on the process used by the traders to form their expectations*. Second, when $e = 1$, the “short run” function F reduces to the “long run” map f , defined by $f(\theta) = F(\theta, 1)$. If we consider a periodic equilibrium with perfect foresight then there are no forecasting errors along the sequence ($e_t = 1$ and $\theta_{t-1}^e = \theta_{t-1}$ for all t), and (5.4) yields the long run relationship $y_t = f(\theta_{t-1})$. *There is thus a non-vanishing “long run statistical relationship” between equilibrium output and real interest rates (or inflation) that holds in the present model along periodic equilibria with perfect foresight, and this relationship does not depend upon the particular cycle under consideration.*

The final step of this investigation is to examine whether a low value of θ_{t-1}^e (i.e. a high anticipated inflation) and/or a low value for the error forecast e_t (i.e. a large unanticipated inflation) is associated with a large equilibrium output y_t , or equivalently, if F is decreasing in both of its arguments. Now consider an increase of θ_{t-1}^e and/or of e_t . It is clear from (5.2) that the equilibrium real balance μ_t goes up, so that the young trader's “wealth” $l_1^* - \mu_t$ goes down, while the old trader's “wealth” $l_2^* + \mu_t$ increases. In view of (5.3) this “redistribution of income” will increase equilibrium output if and only if old traders have a

higher marginal propensity to supply labor, or to consume leisure, provided that leisure is a noninferior good. Formally, we have Assumption (5.i):

ASSUMPTION (5.i): *The functions $c_\tau(a_\tau)$ and $l_\tau(a_\tau)$ are continuously differentiable. The good and leisure are noninferior commodities, that is, $0 < c'_\tau(a_\tau) < 1$ and $-1 < l'_\tau(a_\tau) < 0$ for every $a_\tau \geq 0$, $\tau = 1, 2$. Furthermore, the marginal propensity to consume leisure is higher for an old trader than for a young one, that is $|l'_2(a_2)| > |l'_1(a_1)|$ for all $a_1, a_2 \geq 0$ such that $a_1 + a_2 = l_1^* + l_2^*$.*

It is immediate to verify that under Assumption (5.i), F is a decreasing function of each variable θ and e . Indeed, by differentiation of (5.4),

$$F'_\theta = ez'_2(\theta)[l'_2(l_2^* + ez_2(\theta)) - l'_1(l_1^* - ez_2(\theta))] < 0,$$

$$F'_e = z_2(\theta)[l'_2(l_2^* + ez_2(\theta)) - l'_1(l_1^* - ez_2(\theta))] < 0.$$

Formally, we have obtained Proposition 5.1.

PROPOSITION 5.1: *Assume (1.a), (1.b), (1.c) and define for any positive (θ, e) satisfying $\bar{\theta} \leq \theta \leq z_2^{-1}(l_1^*/e)$,*

$$F(\theta, e) = l_1(l_1^* - ez_2(\theta)) + l_2(l_2^* + ez_2(\theta)).$$

Then for any sequence of temporary competitive equilibrium prices (p_t) satisfying (2.1), (2.2), equilibrium output y_t is linked to anticipated and unanticipated real interest rates by $y_t = F(\theta_{t-1}^e, e_t)$, in which $\theta_{t-1}^e = p_{t-1}/p_t^e$, $\theta_t = p_{t-1}/p_t$ and $e_t = \theta_{t-1}/\theta_{t-1}^e$ for all t . Moreover, under Assumption (5.i), F is continuously differentiable, and $F'_\theta < 0$, $F'_e < 0$.

It is perhaps worth contrasting the results of this section with some *ad hoc* formulations that have been popular recently in the so-called “New Classical” macroeconomic literature. The statistical relationship F closely resembles, if one excepts the fact that our model is highly nonlinear, quite a few log-linear models of the “rational expectations” school. A first important difference is that equilibrium output and real balances, or their deviations from their “permanent” values, which may be viewed as those that correspond to the Golden Rule stationary state $\theta = 1$, depend not only on the “surprise” $e_t = \theta_{t-1}/\theta_{t-1}^e = p_t^e/p_t$ but also on the expected real interest rate $\theta_{t-1}^e = p_{t-1}/p_t^e$. As a result, we are able to obtain a “long run” relationship between equilibrium output and real interest rates (or inflation) described by $f(\theta) = F(\theta, 1)$, even under conditions of complete information and of perfect foresight, by contrast with Lucas [42]. A second important difference is that the “short run” and “long run” relationships F and f involve *equilibrium magnitudes* so that they cannot be interpreted as supply or demand functions as they often are, apparently mistakenly (see, in particular, the so called “Lucas’ supply function”).

6. PROPORTIONAL MONEY TRANSFERS

Periodic competitive equilibria with perfect foresight are Pareto optimal, so it would seem that there is nothing to do about them, except perhaps to reestablish intergenerational equity. The argument is misleading however. Perfect foresight is *very unlikely* to obtain out of long run (here periodic) equilibria. This is especially true in the present model when there are many cyclical equilibria.³⁷ Traders will therefore “learn” and make forecasts that are likely to involve sizeable mistakes and thus large efficiency losses, and the phenomenon will last for a significant amount of time, until the traders’ environment (prices) becomes repetitive enough, i.e. periodic, if it ever does.

There is here an important expectations coordination problem that fails to be solved by the “market.” We show in this section that monetary policy, as exemplified by deterministic proportional money transfers, can solve this coordination problem in this model. Policies of that kind cannot alter, in real terms, the *set* of possible deterministic periodic equilibria with perfect foresight, or their stability properties, by comparison to *laissez-faire*. In this sense, proportional money transfers are “superneutral.” What certain deterministic policy rules can achieve through this channel, however, is to pin down expectations about real interest rates by changing the informational content of prices, to force accordingly the economy to follow a forward perfect foresight path and to converge to any prespecified cycle that is unstable in the b.p.f. dynamics under *laissez-faire*.³⁸ We shall document this claim by looking here at stabilization policies that make the economy converge to the Golden Rule stationary state.

We look first at a trader’s behavior when nominal interest payments are made on money balances. Consider accordingly a young trader who observes the current price $p > 0$, and expects the price $p^e > 0$ to prevail as well as the nominal interest rate $r^e > -1$ to be paid in the future. If $x^e = 1 + r^e > 0$, his problem is to maximize $V_1(a_1) + V_2(a_2)$ with respect to $a_1 \geq 0$, $a_2 \geq 0$, $m \geq 0$ subject to

$$(6.1) \quad pa_1 + m = pl_1^* \quad \text{and} \quad p^ea_2 = p^el_2^* + mx^e.$$

Comparison of this decision problem with (1.5) shows that the optimum excess demands for the good at each age are $z_r(\theta)$, in which $\theta = px^e/p^e$ is again one plus the expected real interest rate, while money demand is

$$(6.2) \quad m^d(p, p^e/x^e) \equiv -pz_1(\theta) \equiv p^ez_2(\theta)/x^e.$$

Competitive equilibrium at date t describes again the equality of supply and demand, that is

$$(6.3) \quad z_1(p_t x_{t+1}^e / p_{t+1}^e) + M_t / p_t = 0,$$

$$(6.4) \quad m^d(p_t, p_{t+1}^e / x_{t+1}^e) = M_t,$$

³⁷ The picture is even worse, since there are many two-states Markov “sunspots” equilibria in the present model as soon as there exists a cycle of period 2 with perfect foresight, as shown by Azariadis and Guesnerie [4].

³⁸ A cycle that is unstable in the b.p.f. dynamics, we recall, is stable in the local *forward* perfect foresight dynamics; see Section 3.

in which p_t is the current price, p_{t+1}^e the expected price, and $r_{t+1}^e = x_{t+1}^e - 1 > -1$ the nominal rate on money balances that is expected for the next date, and M_t is the money supply in period t . As usual, the two equations are equivalent by Walras's Law. The sequence of money supplies M_t is given by

$$(6.5) \quad M_t = x_t M_{t-1}, \quad M_0 \text{ given,}$$

in which $M_{t-1} > 0$ is the beginning of the period (pretransfer) money stock, and $r_t = x_t - 1 > -1$ the nominal interest rate actually paid by the Government at t on money balances. In that set up, the variables actually controlled by the Government are the values of x_t .

Perfect Foresight

Given a sequence of transfers (x_t) , a *periodic equilibrium with perfect foresight* is a sequence of equilibrium prices (p_t) satisfying (6.3), (6.4), (6.5) with $p_{t+1}^e = p_{t+1}$ and $x_{t+1}^e = x_{t+1}$ for all t , such that *real* equilibrium variables (e.g. real interest rates) are periodic. By the same argument that was used to derive (2.3) and (2.4) it is easily seen that such an equilibrium is a periodic solution of the following system:

$$(6.6) \quad z_1(\theta_t) + z_2(\theta_{t-1}) = 0,$$

$$(6.7) \quad M_t / p_t = z_2(\theta_{t-1}),$$

in which $\theta_t = p_t x_{t+1} / p_{t+1}$ for all t , and the sequence of money supplies M_t satisfies (6.5).

The equation (6.6) is identical to (2.3) and thus generates the *same* backward perfect foresight dynamics on real interest rates as in the case of *laissez-faire*, i.e. $\theta_{t-1} = \varphi(\theta_t)$. The b.p.f. dynamics on equilibrium real balances $\mu_t = M_t / p_t$ is also the same, since it is obtained by making the change of variables $\mu_t = z_2(\theta_{t-1})$ in both cases. It is represented here too by $\mu_t = \chi(\mu_{t+1})$. There is therefore a *dichotomy* between real variables and monetary magnitudes in the foregoing system, and *in real terms, the set of deterministic cycles with perfect foresight is independent of the money stocks, and thus of the values of x_t* . The stability or instability of given cycles in the b.p.f. dynamics is not altered either. In this sense, deterministic changes in the rate of growth of the money supply through nominal interest payments are “superneutral.”³⁹

Stabilization Policies

It would be a mistake to conclude from the above remarks that monetary policy is “ineffective”—in the sense that it can influence only nominal magnitudes. *If*

³⁹ The reader will easily verify that the “trade off” between output (or real balances) and anticipated or unanticipated *real* interest rates established in Proposition 5.1 is also independent of the sequence of money stocks.

there are several possible periodic equilibria under *laissez-faire*, nominal interest payments can be used to “guide” the economy toward some of these.

For the sake of simplicity, we focus on stabilization schemes that make the economy converge to the monetary stationary state. The argument can be easily transposed to an arbitrary periodicity. We assume accordingly (1.a), (1.c), (1.d) (the Samuelson case) and look at the case in which the Golden Rule stationary state $\theta = 1$, or $\bar{\mu} = z_2(1)$ is unstable in the b.p.f. dynamics, i.e. $\varphi'(1) < -1$ or $\chi'(\bar{\mu}) < -1$. The instability condition can also be written $z'_1(1) > z'_2(1)$.

We know that in such a case, there exist a nondegenerate cycle of period 2 and a lot of Markov sunspot equilibria, so that the inference problem to be solved by the agents is very complicated. In the absence of any Government's intervention, traders will make significant forecasting errors. The first requirement for a “good” policy rule is thus to pin down expectations. An example is the following one. Suppose that the economy evolves under *laissez-faire* for $t < 0$, but that the Government steps in at $t = 0$ and announces that it will pay interest on money so as to relate the real rate $\theta_t = p_t x_{t+1} / p_{t+1}$ between dates t and $t + 1$ to the rate of inflation between dates $t - 1$ and t , for all $t \geq 0$. Formally⁴⁰

$$(6.8) \quad x_{t+1} = 0 \quad \text{for } t < 0 \text{ and } x_{t+1} = \frac{p_{t+1}}{p_t} \xi\left(\frac{p_t}{p_{t-1}}\right) \quad \text{for } t \geq 0.$$

Remark that the mere announcement of the policy alters the informational content of prices and enables traders to infer the *true* value of θ_t from the observation of p_t / p_{t-1} . So *this policy eliminates also sunspots*. For all $t \geq 0$, the traders' forecasts will obey

$$(6.9) \quad p_t x_{t+1}^e / p_{t+1}^e = \xi(p_t / p_{t-1}) = p_t x_{t+1} / p_{t+1} = \theta_t.$$

Expectations about the price level p_{t+1}^e and the transfer x_{t+1}^e may be wrong, but forecasts of real interest rates will be correct.⁴¹ Plugging (6.9) into the equilibrium condition (6.3) yields for $t = 0$

$$(6.10) \quad z_1(\xi(p_0 / p_{-1})) + M_0 / p_0 = 0$$

which determines p_0 and $\theta_0 = \xi(p_0 / p_{-1})$ as a function of p_{-1} and of M_0 which are taken as given. For $t \geq 1$, using the fact that $M_t / p_t = z_2(\xi(p_{t-1} / p_{t-2}))$ since real interest rates are correctly forecast, we get

$$(6.11) \quad z_1(\xi(p_t / p_{t-1})) + z_2(\xi(p_{t-1} / p_{t-2})) = 0$$

which determines recursively equilibrium prices, and thus equilibrium real rates.

⁴⁰ The policy makes the variable controlled by the Government at date $t + 1$, i.e. x_{t+1} , depend upon the contemporaneously observed price p_{t+1} . The variable x_{t+1} in turn affects the budget constraint of the old traders living at that date. This conforms to the usual Nash equilibrium framework, in which the actions of some agents alter the feasible set of strategies of others. A similar phenomenon arises in the Arrow–Debreu model, because firms' profits depend on prices whereas profits influence consumers' wealth.

⁴¹ It is essential to get this result that the policy rule links the *real* rate of interest to previously observed variables. By contrast, there would be no hope of getting this result if x_{t+1} was *predetermined* at each date, i.e., if it depended upon variables up to date t , but not on p_{t+1} .

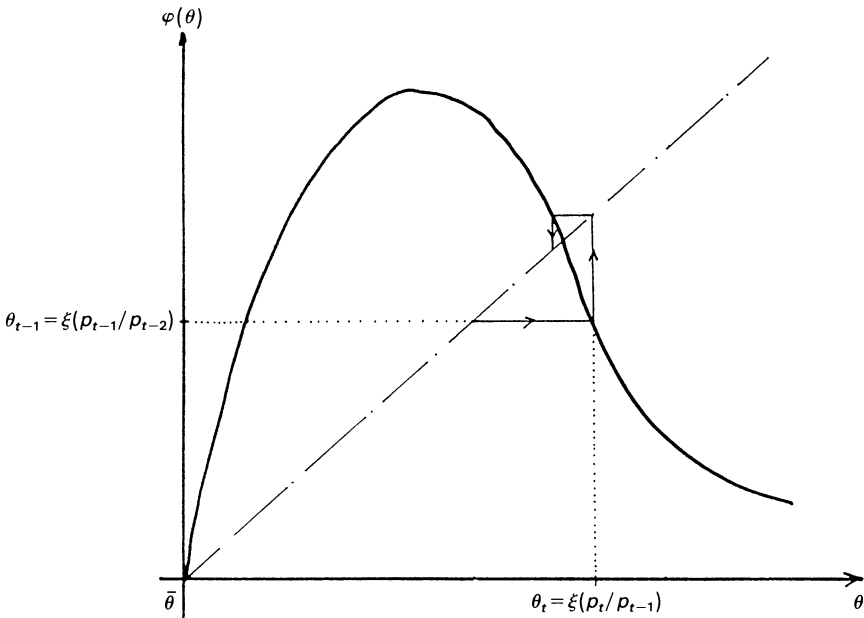


FIGURE 5

Since real rates are given by $\theta_t = \xi(p_t/p_{t-1})$, (6.11) apparently reduces to (6.6). There is an important difference, however. In (6.6), the range of variation of θ_t is not limited. As a consequence, there will be typically several values of θ_t solving (6.6) given θ_{t-1} (see Figure 5). By contrast the range of variation of θ_t is limited in (6.11) by the range of the policy function ξ . Thus by restricting appropriately the range of ξ , one can force θ_t to stay in a neighborhood of the stationary state $\theta = 1$. Then, as shown in Figure 5, given $\theta_{t-1} = \xi(p_{t-1}/p_{t-2})$, the value of $\theta_t = \xi(p_t/p_{t-1})$ solving (6.11) is uniquely defined. The sequence θ_t will follow the *local* forward perfect foresight dynamics around $\theta = 1$, and will actually converge to this value since the Golden Rule stationary state has been assumed to be unstable in the *backward* perfect foresight dynamics.

We end this Section with a precise proof of this heuristic statement. We assume:

(6.12) ξ is continuous, decreasing, and maps $(0, +\infty)$ onto some interval

$$I = (1 - \varepsilon, 1 + \varepsilon). \text{ Moreover, } \xi(x^*) = 1.$$

In this formulation, $x^* - 1 > -1$ is to be interpreted as a target inflation rate that is to be achieved in the long run together with the goal $\theta = 1$, hence the relation $\xi(x^*) = 1$. The fact that ξ is decreasing means that the Government wishes to slow down the real rate of growth of the money supply, i.e. $p_t x_{t+1}/p_{t+1}$, when inflation has been high in the immediate past (countercyclical policy). We have then Proposition 6.1.

PROPOSITION 6.1: *Assume (1.a), (1.c), (1.d) and that the Golden Rule stationary state is unstable in the b.p.f. dynamics, i.e. $\varphi'(1) < -1$, or $z'_1(1) > z'_2(1)$. Suppose that the Government announces the policy rule (6.8) and that it satisfies (6.12). Then if ε is small enough, the economy follows a forward perfect foresight path that is uniquely defined by (6.9), (6.10), (6.11), given M_0 and p_{-1} , and converges to the stationary state $\theta = 1$, while the rate of inflation and the nominal interest rate tend to $x^* - 1$.*

The proof of this claim is simple. If ε is small enough, so that $z'_1(\xi(v)) > 0$ for all v , then the left-hand side of (6.10) is, given p_{-1} and M_0 , a decreasing function of p_0 , which tends to $+\infty$ when p_0 goes to 0 and becomes negative for p_0 large enough. Thus (6.10) has a unique solution in p_0 , which yields $\theta_0 = \xi(p_0/p_{-1})$.

Given θ_0 in $I = (1 - \varepsilon, 1 + \varepsilon)$, a sequence $\theta_t = \xi(p_t/p_{t-1})$ verifies (6.11) for all $t \geq 1$ if and only if it satisfies $\theta_{t-1} = \varphi(\theta_t)$ and θ_t belongs to I for all t . Now since $\varphi'(1) < -1$, and for ε small enough, the restriction of φ to the interval I , i.e. $\varphi|_I$, is invertible. Its inverse $(\varphi|_I)^{-1}$ maps the open interval $I' = \varphi(I)$ onto I , is continuously differentiable, decreasing, and is a contraction. Given θ_0 in I , the sequence we are looking for is defined recursively by $\theta_t = (\varphi|_I)^{-1}(\theta_{t-1})$ and it converges to $\theta = 1$. Moreover, since the range of ξ is I , for each element θ_t of this sequence, there exists a unique p_t solving $\theta_t = \xi(p_t/p_{t-1})$ given p_{t-1} . In fact the ratio p_t/p_{t-1} as well as x_t converge to x^* when t tends to $+\infty$. This completes the proof of the Proposition.^{42,43}

7. CONCLUSION

The model that we have analyzed in this paper is obviously too rudimentary to enable us to draw very general conclusions. Yet, the study of this particular example is suggestive and shows that it is possible to construct plausible models in which endogenous and significant fluctuations obtain that are not caused by exogenous shocks nor by policy. Furthermore a properly designed countercyclical policy can stabilize these autonomous business cycles. Such results are manifestly at variance with the conclusions that yield usually currently fashionable Classical “loglinear” macroeconomic models. It remains to be seen whether or not these

⁴² One may remark that each equilibrium price p_t determined by (6.10) and (6.11) is stable in any Walrasian tatonnement process in which the price of the good responds positively at date t to an excess demand at that date, given past prices. This comes from the fact that the left-hand side of (6.10) or of (6.11) is a decreasing function of p_0 , or of p_t . The dynamics of the economy, viewed as a sequence of temporary competitive equilibria, are well defined.

⁴³ It should be emphasized that the “superneutrality” of money injection through proportional money transfers no longer holds if money is injected into the economy through lump sum transfers or Government’s purchases of goods. Such policies can thus create new cycles as well as stabilize them. The distinction between the effects of a variation of the rate of growth of the money supply through nominal interest payments and through lump sum transfers is in particular at the heart of Friedman’s theory of the optimum quantity of money [23, Ch. 1]. See also Grandmont and Younès [37] and the discussion in Grandmont [33, Ch. 1, Section 7]. This distinction has been used again recently in the perspective of stabilization policy by Azariadis and Balasko [3], Stiglitz [55], Woodford [60]. The effects of a change of the money supply through the Government’s purchases has been studied recently in a (log-linear) version of the overlapping generation model by S. Fisher [21].

results carry over to more general and more realistic structures. They suggest at any rate that economic theorists should perhaps look more closely at the sort of mechanisms that may be responsible for significant nonlinearities in the economic system if they wish to have a proper foundation upon which to build a sound business cycle theory.

Among the issues that appear worthwhile to investigate, one may mention the following:

1. One important feature of the model that has been studied is the fact—that seems likely to obtain as well within more sophisticated frameworks—that cycles of different periods typically coexist. This raises, as we have seen, the question of the stability of a given cycle, and our analysis showed that any serious analysis of this issue must take into account the fact that traders are learning on the transition path. We have in fact touched upon this problem only incidentally in Section 3, when establishing a relationship between stability with learning and stability under the perfect foresight backward dynamics. A deeper examination of this issue, for instance by adapting and simplifying the methods of Fuchs and Laroque [29], would be useful.

2. Although we have deliberately ignored exogenous shocks in the present analysis in order to focus attention on autonomous business cycles, it is clear that (random or deterministic) shocks that result from the economy's environment and/or from the Government's policy, as well as incomplete and asymmetric information, play *some* role in the generation of modern economic fluctuations, although their importance may not be as great as some "New Classical macroeconomists" would like to believe. The integration of such exogenous shocks, for instance along the lines of Lucas's model [42], would be interesting as it would allow making precise how they interact with the endogenous mechanisms in the generation of economic fluctuations.

3. A related issue concerns the fact already mentioned in Section 6 that deterministic economic policies through lump sum money transfers, income taxes, Government purchases of goods, open market operations are able to generate economic fluctuations as well as to stabilize them. It would be important to proceed to a systematic study of the existence and the properties of the cycles that may emerge in the present model when the Government is implementing a given policy through these alternative channels.

4. As we have said, the central mechanism through which endogenous business cycles emerge in the present model is the conflict between the intertemporal substitution effect and the wealth effect that are associated with a variation of the real rate of interest. An interesting study would be to extend the analysis to a more general framework, in which, for instance, traders would live an arbitrary (finite or infinite) number of periods. Another topic of particular interest is to discover other mechanisms, for instance associated with particular features of the production technology, that yield nonlinearities sufficient to generate endogenous business cycles.

5. The whole analysis has been carried out under the *ad hoc* assumption that markets clear at every moment in the Walrasian sense. An important topic for

future research would be to relax this somewhat unrealistic assumption, for instance by introducing imperfect competition in the model, and to look for the kind of mechanisms that may give rise to endogenous economic fluctuations. In a Walrasian framework, one may get economic fluctuations only through a variation of relative prices, that is (in a macroeconomic context involving only one good as in the present model) only through a variation of the real interest rate. By contrast, once imperfect competition is taken into account, *quantity* adjustments, and thus something like multiplier or accelerator effects, have to play a role. One may then hope to get a better foundation upon which to build a sound Keynesian, or more precisely non-Walrasian, business cycle theory.

CEPREMAP

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APPENDIX

A. STABILITY AND LEARNING

The purpose of this Section is to prove Proposition 3.2. We assume therefore (1.a), (1.c), (3.f), (3.g), (3.h), and consider a cycle $(p_k^*, \dots, p_1^*)$ of the map Φ . Assume that this cycle is Φ -stable, i.e. $\prod_1^k |\Phi'(p_i^*)| < 1$. We wish to show that this cycle is also stable in the forward dynamics implied by (3.6). Or more precisely, if $(q_1^*, \dots, q_k^*)$ is the associated periodic orbit of the map $\tilde{W}$ defined in (3.7), with

$$q_i^* = (p_i^*, \dots, p_1^*, p_k^*, \dots, p_1^*, \dots),$$

we wish to show that the eigenvalues of the Jacobian matrix

$$D\tilde{W}^k(q_1^*) = D\tilde{W}(q_k^*) \dots D\tilde{W}(q_1^*)$$

all have a modulus less than 1 (see 3.9). Clearly, it is enough to prove the proposition for $\varepsilon = 0$, for in that case it would still be true, by continuity, for ε small enough.

The b.p.f. dynamics on prices $p_t = \Phi(p_{t+1})$ is obtained by setting $p_{t+1}^e = p_{t+1}$ in (2.1), (2.2). This system is thus equivalent to $p_t = \Phi(p_{t+1}^e)$, and the temporary competitive equilibrium map W is defined implicitly by

$$(A.1) \quad W(p_{t-1}, \dots, p_{t-T}) \equiv \Phi[\psi(W(p_{t-1}, \dots, p_{t-T}), p_{t-1}, \dots, p_{t-T})].$$

As an incidental remark, we note that we may assume without any loss of generality, for the purpose of the present proof, that T is a multiple of k , that is $T = mk$ for some integer $m \geq 1$. Indeed it is always possible to take into account an arbitrary number of additional past prices as fictitious arguments of the expectation function ψ , without violating (3.f), (3.g), (3.h).

Let $W'_j(q_j^*)$ be the partial derivative of W with respect to p_{t-j} for $j = 1, \dots, T$, evaluated at $(p_{t-1}, \dots, p_{t-T}) = q_j^*$. By differentiating (A.1) and by using Assumptions (3.g) and (3.h) with $\varepsilon = 0$, we get the following:⁴⁴

$$(A.2) \quad \text{For every } i = 1, \dots, k, \text{ one has } W'_j(q_i^*) = 0 \text{ for all } j = 1, \dots, T (=mk), \text{ except for those of the form } j = nk - 1, \text{ for which one has } W'_j(q_i^*) = \Phi'(p_{i+2}^*)\psi'_j(p_{i+1}^*, q_i^*) \text{ (here the integer } n \text{ runs from 2 to } m+1 \text{ when } k=1, \text{ and from 1 to } m \text{ when } k \geq 2). \text{ Accordingly, for all such } n, W'_{nk-1}(q_i^*) \text{ has the same sign as } \Phi'(p_{i+2}^*), \text{ and one has } \sum_n W'_{nk-1}(q_i^*) = \Phi'(p_{i+2}^*).$$

Our next task is to evaluate the Jacobian matrix $D\tilde{W}^k(q_1^*)$. We first remark that in view of (3.7), one has for every $q = (q_1, \dots, q_T)$ in the interior of $\mathbb{R}_+^T$,

$$(A.3) \quad \tilde{W}(q) = (W(q), q_1, \dots, q_{T-1}).$$

⁴⁴ We use the convention $p_{k+h}^* = p_h^*$ and the notation $\psi'_j(p_{i+1}^*, q_i^*)$ stands for $\psi'_j(p_{i+1}^*, p_i^*, \dots, p_1^*, p_k^*, \dots)$.

Thus we get by differentiation of (A.3),

$$D\tilde{W}(q) = \begin{bmatrix} W'_1(q) & W'_2(q) & W'_3(q) & \cdots & W'_{T-1}(q) & W'_T(q) \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}.$$

If $k = 1$ (hence $T = m$), we are done since it suffices to make q equal to the stationary vector q^* in the above matrix. When $k \geq 2$, one computes $D\tilde{W}^k(q^*)$ with the help of the above matrix by using the fact that $D\tilde{W}^k(q_1^*) = D\tilde{W}(q_k^*) \cdots D\tilde{W}(q_1^*)$ and by using the restrictions stated in (A.2). The result of these (tedious) computations is that one may partition, for every $k \geq 1$, the Jacobian matrix $D\tilde{W}^k(q_1^*)$ into m^2 submatrices of dimensions (k, k) as follows:

$$D\tilde{W}^k(q_1^*) = \begin{bmatrix} A_1 & A_2 & A_3 & \cdots & A_{m-2} & A_{m-1} & A_m \\ I & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & I & 0 & \cdots & 0 & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & \cdots & I & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & I & 0 \end{bmatrix},$$

in which I is the (k, k) unit matrix, 0 is a (k, k) matrix of which the elements are all 0 , and for each $n = 1, \dots, m$, the matrix A_n is given by

$$A_n = \begin{bmatrix} 0 & 0 & 0 & \cdots & w_{1n}w_{k1} & w_{k,n+1} \\ w_{k-1,n} & 0 & 0 & \cdots & 0 & 0 \\ 0 & w_{k-2,n} & 0 & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & w_{1n} & 0 \end{bmatrix}$$

with $w_{in} = W'_{nk-1}(q_i^*)$ for $i = 1, \dots, k$ and $n = 1, \dots, m$, and with the additional convention that $w_{k,m+1} = W'_{(m+1)k-1}(q_k^*)$ vanishes when $k \geq 2$. When $k = 1$, each matrix A_n reduces then to its top right element $w_{1,n+1}$, which is equal to $W'_n(q^*)$ with q^* being the fixed point of $\tilde{W}$ under consideration.

Note that with this notation, (A.2) means that given $i = 1, \dots, k$ all w_{in} have the same sign and $\sum_n w_{in} = \Phi'(p_{i+2}^*)$ (here again n runs from 1 to m when $k \geq 2$ and from 2 to $m+1$ when $k = 1$).

Our purpose, we recall, is to show that $|\Phi'(p_1^*) \cdots \Phi'(p_k^*)| < 1$ implies that all eigenvalues of $D\tilde{W}^k(q_1^*)$ have a modulus less than 1 . These eigenvalues are the solutions of the characteristic equation

$$\det(D\tilde{W}^k(q_1^*) - \lambda I) = 0$$

in which I is this time the unit (T, T) matrix. It is known that given the specific form of the Jacobian matrix $D\tilde{W}^k(q_1^*)$ shown above, this characteristic equation may be written equivalently $\det(B(\lambda)) = 0$, in which $B(\lambda)$ is the (k, k) matrix that is equal to⁴⁵

$$B(\lambda) = \lambda^m I - \sum_{n=1}^m \lambda^{m-n} A_n$$

(here I denotes the (k, k) unit matrix). If $k = 1$, the characteristic equation becomes

$$(A.4) \quad 0 = \lambda^m - \sum_{n=1}^m w_{1,n+1} \lambda^{m-n}.$$

⁴⁵ See Wilkinson [59, pp. 33-34].

If $k \geq 2$, we obtain, by employing the expressions of the matrices A_n given above,

$$B(\lambda) = \begin{bmatrix} \lambda^m & 0 & 0 & \cdots & -w_{k1}b_1(\lambda) & -c_k(\lambda) \\ -b_{k-1}(\lambda) & \lambda^m & 0 & \cdots & 0 & 0 \\ 0 & -b_{k-2}(\lambda) & \lambda^m & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & \lambda^m & 0 \\ 0 & 0 & 0 & \cdots & -b_1(\lambda) & \lambda^m \end{bmatrix}$$

with, for $i = 1, \dots, k$,

$$b_i(\lambda) = \sum_{n=1}^m w_{in} \lambda^{m-n}$$

and

$$c_k(\lambda) = \sum_{n=1}^m w_{k,n+1} \lambda^{m-n}.$$

The expansion of the determinant yields

$$\det(B(\lambda)) = \lambda^{mk} - b_{k-1}(\lambda) \cdots b_1(\lambda)(w_{k1}\lambda^m + c_k(\lambda)).$$

It we recall that $w_{k,m+1} = 0$ whenever $k \geq 2$, we see that

$$\det(B(\lambda)) = \lambda^{mk} - \lambda b_1(\lambda) \cdots b_k(\lambda)$$

when $k \geq 2$; the characteristic equation becomes therefore

$$(A.5) \quad 0 = \lambda^{mk} - \lambda \left(\sum_{n=1}^m w_{1n} \lambda^{m-n} \right) \cdots \left(\sum_{n=1}^m w_{kn} \lambda^{m-n} \right).$$

The last step of the proof is to show that $|\Phi'(p_1^*) \cdots \Phi'(p_k^*)| < 1$ implies that all solutions of (A.4) when $k = 1$, or of (A.5) when $k \geq 2$ have a modulus less than 1. Consider any complex number λ with $|\lambda| \geq 1$. If $k = 1$, let p^* be the stationary price and $q^* = (p^*, p^*, \dots)$. Since all $w_{1,n+1} = W'_n(q^*)$ have the same sign and sum to $\Phi'(p^*)$, we have

$$\begin{aligned} \left| \sum_{n=1}^m w_{1,n+1} \lambda^{m-n} \right| &\leq \sum_{n=1}^m |w_{1,n+1}| |\lambda|^{m-n} \\ &\leq \sum_{n=1}^m |w_{1,n+1}| |\lambda|^m = |\Phi'(p^*)| |\lambda|^m \\ &< |\lambda|^m \end{aligned}$$

and all roots of (A.4) must lie in the interior of the unit circle. When $k \geq 2$, the reasoning for (A.5) is similar. Given $i = 1, \dots, k$, all w_{in} have the same sign and sum to $\Phi'(p_{i+2}^*)$. Thus if $|\lambda| \geq 1$,

$$\left| \sum_{n=1}^m w_{in} \lambda^{m-n} \right| \leq |\Phi'(p_{i+2}^*)| |\lambda|^{m-1}.$$

Therefore

$$\begin{aligned} \left| \lambda \prod_{i=1}^k \left(\sum_{n=1}^m w_{in} \lambda^{m-n} \right) \right| &\leq |\Phi'(p_1^*) \cdots \Phi'(p_k^*)| |\lambda|^{(m-1)k+1} \\ &< |\lambda|^{mk} \end{aligned}$$

and again all roots of (A.5) must have a modulus less than 1. This completes the proof of Proposition 3.2.

B. EXISTENCE, STABILITY AND BIFURCATION OF PERIODIC EQUILIBRIA

The aim of this Section is to prove some of the claims made in Section 4, namely Theorems 4.3, 4.5, and 4.7. The arguments will be direct applications of the results reported in Collet and Eckmann [16], as reviewed in Grandmont [34].

PROOF OF THEOREM 4.3: If $\varphi(\theta) \leq 1$ for $\theta \leq 1$, then by the same argument as in Lemma 4.2, the only cycles of φ are the two fixed points $\theta = \bar{\theta}$ and $\theta = 1$. In fact, one has $\lim_{j \rightarrow \infty} \varphi^j(\theta) = 1$ for all $\theta > \bar{\theta}$. If $\varphi(\theta) > 1$ for some $\theta < 1$, let θ_0 be a point of $(\bar{\theta}, 1)$ such that φ reaches its maximum on the interval $[\bar{\theta}, 1]$ at θ_0 . Then φ maps the interval $I = [\bar{\theta}, \varphi(\theta_0)]$ into itself, and for every $\theta > \varphi(\theta_0)$, there exists j such that $\varphi^j(\theta)$ belongs to I . Thus all cycles of φ lie in I . The result then follows by applying Sarkovskii's Theorem (Grandmont [34, Theorem 1]) to the restriction of φ to I . Q.E.D.

PROOF OF THEOREM 4.5: Remark first that $\varphi(\theta)$ belongs to the interval $J = [\bar{\theta}, \varphi(\theta^*)]$ whenever $\theta \geq \bar{\theta}$. Then if φ has a negative Schwarzian derivative on that interval, Theorem 2 in Grandmont [34] applies to the restriction of φ to J . Assume now that one topological conjugate of φ , i.e. $\tilde{\varphi} = h_0 \varphi_0 h^{-1}$, has a negative Schwarzian derivative on $h(J) = [h(\bar{\theta}), h(\varphi(\theta^*))]$. One may assume without loss of generality that h is increasing; if not, make a linear change of variable that reverses orientation. Then Theorem 2 in Grandmont [34] applies to the restriction $\tilde{\varphi}$ to $h(J)$. Q.E.D.

PROOF OF THEOREM 4.7: Here again, we may focus attention on the interval $A_\lambda = [\bar{\theta}_\lambda, \varphi_\lambda(\theta_\lambda^*)]$ for each λ . The result is then a direct application of Theorems 7 and 9 in Grandmont [34]. Indeed it is routine to verify that if one looks at the restrictions of φ_λ to A_λ , one gets a full one-parameter family of C^1 -unimodal maps that satisfies S1, S2, S3, S4" for each λ , as these conditions are defined in Grandmont [34]. As for the condition $\varphi_\lambda''(\theta_\lambda^*) < 0$ it is not difficult to verify that it is satisfied if and only if $\chi''(\mu^*) < 0$. Dropping the index λ for simplicity, we get by differentiating twice $v_1(\chi(\mu)) = v_2(\mu)$ and by taking into account $\chi'(\mu^*) = 0$,

$$\chi''(\mu^*) = v_2''(\mu^*) / v_1'(\chi(\mu^*)).$$

Since $v_1'(\mu) > 0$ for all μ , $\chi''(\mu^*) < 0$ is equivalent to $v_2''(\mu^*) < 0$. Now by differentiation of $v_2(\mu) = \mu V_2(I_2^* + \mu)$, we obtain

$$v_2'(\mu) = V_2(I_2^* + \mu)\rho(\mu)$$

where

$$\rho(\mu) = 1 - \frac{\mu}{I_2^* + \mu} R_2(I_2^* + \mu).$$

Differentiating again,

$$v_2''(\mu^*) = V_2(I_2^* + \mu^*)\rho'(\mu^*) < 0$$

since $\rho(\mu^*) = 0$ and $\rho'(\mu) < 0$ under Assumption (1.e).

Q.E.D.

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